

# SIGN PATTERNS OF BORWEIN PRODUCTS BY EXPLICIT CIRCLE-METHOD AND SADDLE-POINT CERTIFICATES: THE CUBIC BORWEIN CONJECTURE AND THE THIRD BORWEIN CONJECTURE

JAIDEEP SAI PADHI

ABSTRACT. For a prime  $p$  put  $P_{p,n}(q) = \prod_{k \leq pn, p \nmid k} (1 - q^k)$ . We develop a method for proving sign patterns of the coefficients of  $P_{p,n}(q)^\delta$  modulo  $p$ , and use it to settle two open cases. First, we prove the remaining residue class of the cubic Borwein conjecture:  $[q^m]P_{3,n}(q)^3 \leq 0$  for all  $n \geq 1$  and all  $m \equiv 2 \pmod{3}$ ; together with the work of Wang and Krattenthaler this proves the conjecture. Second, we prove the Third Borwein conjecture: the coefficients of  $P_{5,n}(q) = (q; q)_{5n}/(q^5; q^5)_n$  are  $\geq 0$  at exponents divisible by 5 and  $\leq 0$  otherwise.

The method factors  $P_{p,n}^\delta = G_p^\delta E_n$  with  $G_p = (q; q)_\infty / (q^p; q^p)_\infty$  and a tail  $E_n$  with non-negative coefficients, uses an explicit circle-method expansion of  $G_p^\delta$  with a fully proved error constant, and splits the coefficient range into an exact range, a band, a Laplace (saddle-point) region and a centre. Each analytic region is reduced to finitely many inequalities in a few real parameters, which are verified in ball arithmetic uniformly in  $n$ . For  $p = 5$  new difficulties arise: two residue classes in which the main term vanishes, which require re-centring the saddle point on the thin factor, and a centre region that requires an off-peak bound for  $|P_{5,n}|$  on the whole circle.

*Both proofs are computer-assisted.* All programs, their logs, and the lines each log must contain are published with the paper and are described in §5.

## CONTENTS

|                                                    |    |
|----------------------------------------------------|----|
| 1. Introduction                                    | 1  |
| 2. The general framework                           | 3  |
| 3. The cubic case: the class $m \equiv 2 \pmod{3}$ | 6  |
| 4. The Third Borwein conjecture                    | 12 |
| 5. Computational verification                      | 29 |
| Appendix A. Verification and audit record          | 32 |
| Appendix B. Standard facts and elementary lemmas   | 34 |
| References                                         | 35 |

## 1. INTRODUCTION

For a prime  $p$  and  $n \geq 1$  put

$$P_{p,n}(q) = \prod_{\substack{k \leq pn \\ p \nmid k}} (1 - q^k) = \frac{(q; q)_{pn}}{(q^p; q^p)_n}.$$

P. Borwein observed, and Andrews [1] made widely known, that the coefficients of such products and of their small powers appear to have a sign pattern that depends only on the exponent modulo

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$p$ . The three conjectures that concern us are the following. Write  $c(m)$  for the coefficient of  $q^m$  in the polynomial under discussion.

- (1) (*First Borwein conjecture*.) For  $P_{3,n}$ :  $c(m) \geq 0$  if  $m \equiv 0$  and  $c(m) \leq 0$  if  $m \equiv 1, 2 \pmod{3}$ .
- (2) (*Cubic Borwein conjecture*.) The same pattern  $+$   $-$   $-$  holds for  $P_{3,n}^3$ .
- (3) (*Third Borwein conjecture*.) For  $P_{5,n}$ :  $c(m) \geq 0$  if  $5 \mid m$  and  $c(m) \leq 0$  otherwise.

There is also a second Borwein conjecture, for  $P_{3,n}^2$ . (The names First, Second and Third follow [14, 3]; Andrews [1] discusses Borwein's conjectures for  $p = 3$  and  $p = 5$ .) The first conjecture was proved by Wang [13], by an analytic (saddle-point) method. Wang and Krattenthaler [14] gave a new proof of the first conjecture and proved the second by an asymptotic approach to Borwein-type sign patterns; they also proved “two thirds” of the cubic conjecture (see also [3, §1]), namely the class  $m \equiv 0 \pmod{3}$  and, in the range  $m \leq \frac{9}{2}n^2$  (equivalently, by the palindromic symmetry of  $P_{3,n}^3$ , the complementary halves), the class  $m \equiv 1 \pmod{3}$ , and explain [14, §11] why the class  $m \equiv 2$  resists their method. The Third Borwein conjecture is listed as open in the recent literature [3, Conj. 1.3], where it is noted that [14] provides ideas for “three fifths” of a proof; we are not aware of a proof in the literature.

### 1.1. Main results.

**Conjecture 1.1** (cubic Borwein conjecture). Let  $c_n(m) = [q^m]P_{3,n}(q)^3$ . For all  $n \geq 1$  and  $0 \leq m \leq \frac{9}{2}n^2$ :  $c_n(m) \geq 0$  if  $m \equiv 0$ , and  $c_n(m) \leq 0$  if  $m \equiv 1, 2 \pmod{3}$ .

**Theorem A** (the remaining class of the cubic conjecture). *For all  $n \geq 1$  and all  $m \equiv 2 \pmod{3}$  with  $m \leq \frac{9}{2}n^2$  we have  $c_n(m) = 0$  if  $m < 3n$  and  $c_n(m) < 0$  if  $m \geq 3n$ . Together with [14] this proves Conjecture 1.1.*

**Conjecture 1.2** (Third Borwein conjecture). Let  $c_n(m) = [q^m]P_{5,n}(q)$ . For every  $n \geq 1$  and every  $m$ ,  $c_n(m) \geq 0$  if  $m \equiv 0 \pmod{5}$  and  $c_n(m) \leq 0$  if  $m \not\equiv 0 \pmod{5}$ .

**Theorem B.** *Conjecture 1.2 holds.*

Theorem A is Theorem 3.1 and Theorem B is Theorem 4.1. Both are computer-assisted. Theorem A needs only the one residue class  $m \equiv 2 \pmod{3}$ ; Theorem B is proved for all five classes modulo 5.

**1.2. Why the remaining cases are hard.** Both polynomials factor as  $P_{p,n}^\delta = G_p^\delta E_n$  (§2), where  $G_p = (q; q)_\infty / (q^p; q^p)_\infty$  and  $E_n$  is a tail product with non-negative coefficients. The circle method gives the coefficients of  $G_p^\delta$  with a main term from the cusps of denominator  $p$ , and the tail  $E_n$  is a small perturbation of 1 for exponents  $m$  below  $N = pn + 1$ . The difficulty is that in the classes we must treat, the main term of  $G_p^\delta$  vanishes:

- for  $p = 3$ ,  $\delta = 3$ , all coefficients of  $G_3^3$  in the class  $m \equiv 2$  vanish (Lemma 2.3);
- for  $p = 5$ ,  $\delta = 1$ , all coefficients of  $G_5$  in the classes  $m \equiv 3, 4$  vanish (Lemma 2.3).

In these “thin” classes  $c_n(m)$  is produced entirely by the tail  $E_n$ , and the leading saddle-point contribution cancels. The sign then comes from a secondary term that is exponentially smaller than the naive main term.

**1.3. Overview of the method.** The coefficient range is split into four regions (§2.4).

- (1) *Exact range.* Small  $n$  are handled by exact integer computation.
- (2) *Band.* For  $m$  below a small multiple of  $N$  only a few parts of  $E_n$  contribute, and  $c_n(m)$  is a short, explicit combination of coefficients of  $G_p^\delta$ , controlled by exact values and by the circle-method expansion.

- (3) *Laplace region.* Here an exact Bessel–Laplace (or Poisson) representation of the main term is combined with the closed form of  $\log E_n$  at a twisted point, and a saddle-point analysis is carried out in scaled variables.
- (4) *Centre.* Near the middle of the coefficient range, Cauchy’s formula for the polynomial is evaluated directly by steepest descent through the saddles near the primitive  $p$ -th roots of unity.

The method was developed on the cubic case. For  $p = 5$  three new difficulties arise:

- *two* vanishing classes instead of one;
- in the Laplace region for the classes 3, 4 the surviving term carries a factor  $e^{-5nu}$  which is of size  $e^{O(N)}$  in the exponent; the saddle must be re-centred on this “thin factor” (§4.5.4);
- in the centre, the contribution of the circle away from the four peaks must be bounded by an explicit *off-peak lemma* for  $|P_{5,n}|$  (§4.6.2), which is then combined with the near-peak analysis box by box.

The two proofs share the method, not their results: the cubic theorem is not an ingredient of the proof for  $p = 5$ , and conversely.

**1.4. What is computer-assisted, and how it is verified.** Each analytic region is reduced to finitely many inequalities between explicit functions of one to three real parameters (a scaled saddle point, the reciprocal  $1/N$ , an angle), and these are verified on coverings of the parameter domains by balls in Arb ball arithmetic [7]. Uniformity in  $n$  is obtained in two ways: the scaled integrands are analytic in  $1/N$  at 0, so one ball covers all large  $N$ ; and every term that depends on  $n$  is shown, by an asserted inequality, to be non-increasing in  $n$ . Exact ranges are computed with FLINT [12] in  $\mathbb{Z}[q]$ .

Throughout, a statement proved by a computation is stated as a lemma or a certificate, followed by a line *Certificate:* naming the program and its log. Section 5 lists, for every certificate, the command, the log and the line the log must contain; Appendix A summarises the independent audits of the proof of Theorem B. The trust base consists of the written arguments of this paper, the standard facts of Appendix B, and the correctness of Arb and FLINT.

**1.5. Organisation.** Section 2 sets up the common framework and notation. Section 3 proves Theorem A and Section 4 proves Theorem B. Section 5 describes the certificates.

## 2. THE GENERAL FRAMEWORK

Throughout,  $(p, \delta)$  is one of the two cases

$$(p, \delta) = (3, 3) \quad (\text{Section 3}), \quad (p, \delta) = (5, 1) \quad (\text{Section 4}),$$

and  $c_n(m) = [q^m]P_{p,n}(q)^\delta$ .

**2.1. Notation.** Table 1 fixes the symbols shared by the two parts. Where the two source proofs used one letter for different objects, one of them has been renamed; symbols that are local to a single subsection are defined where they are used and are not listed.

**2.2. Factorisation, palindromy and vanishing classes.**

**Lemma 2.1** (factorisation).  $P_{p,n}^\delta = G_p^\delta E_n$ . The series  $E_n$  has non-negative coefficients,  $e(0) = 1$  and  $e(j) = 0$  for  $0 < j < N$ . Consequently

$$c_n(m) = \sum_{j \geq 0} e(j) s(m-j) = s(m) + \sum_{j \geq N} e(j) s(m-j).$$

*Proof.*  $G_p = \prod_{p \nmid k} (1 - q^k)$ . Split this product at  $k = pn$ : the factor over  $k > pn$  is  $E_n^{-1/\delta}$ , and  $E_n$  is a product of  $(\delta$ -th powers of) geometric series in  $q^k$  with  $k \geq pn + 1 = N$ .  $\square$

| symbol                  | meaning                                                                                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $P_{p,n}; P_n$          | $\prod_{k \leq pn, p \nmid k} (1 - q^k)$ ; in §3 and §4 $p$ is fixed and we write $P_n$                                                                                                   |
| $c_n(m)$                | $[q^m]P_{p,n}^\delta$ ; $\deg P_{p,n}^\delta = \delta(p-1)pn^2/2$ ( $9n^2$ for $p=3$ , $10n^2$ for $p=5$ )                                                                                |
| $G_p; s(i)$             | $(q; q)_\infty / (q^p; q^p)_\infty = \prod_{p \nmid k} (1 - q^k)$ ; $s(i) = [q^i]G_p^\delta$                                                                                              |
| $E_n; e(j)$             | $\prod_{k > pn, p \nmid k} (1 - q^k)^{-\delta}$ , so that $P_{p,n}^\delta = G_p^\delta E_n$ ; $e(j) = [q^j]E_n \geq 0$                                                                    |
| $N$                     | $pn+1$ (the smallest part of $E_n$ ); in §4.2 only, $N$ is the Farey order                                                                                                                |
| $\omega_p$              | $e^{2\pi i/p}$ (so $\omega_3 = \omega$ of §3, $\omega_5$ of §4)                                                                                                                           |
| $B, C$                  | $B = \delta(p-1)/12$ ( $= \frac{1}{2}$ resp. $\frac{1}{3}$ ), $C = C(i) = 2i - B$                                                                                                         |
| $\mathcal{P}_k(i)$      | $\frac{2\pi}{k} \sqrt{B/C} I_1(\frac{2\pi}{k} \sqrt{BC})$ for $i \geq 1$ ; $\mathcal{P}_k(i) = 0$ for $i \leq 0$                                                                          |
| $a_k; a_p$              | $a_k = 2\pi^2 B/k^2$ ; so $a_3 = \pi^2/9$ (cubic), $a_5 = 2\pi^2/75$ (Third), and $\mathcal{P}_p(i) = \sqrt{a_p/x} I_1(2\sqrt{a_p x})$ with $x = i - B/2$                                 |
| $\mathfrak{a}(i)$       | the amplitude of the $k=p$ main term: $\mathfrak{a}(i) = \sqrt{3}\chi(i)$ for $p=3$ ; $\mathfrak{a}(i) = 2\cos(2\pi i/5 + \pi/5) + 2\cos(4\pi i/5)$ for $p=5$                             |
| $K(i)$                  | $\lfloor \sqrt{4\pi i} \rfloor + 2$                                                                                                                                                       |
| $E_{\text{con}}$        | error constant of the expansion: 50.1501 ( $p=3$ ), 27.4636 ( $p=5$ )                                                                                                                     |
| $\mu$                   | $m/\deg P_{p,n}^\delta$ , so the half-range is $\mu \leq \frac{1}{2}$ (used in §4); in §3 the centre is parametrised by $\bar{\mu} = m/N^2$ and $\tilde{\mu} = (m - N - \frac{1}{4})/N^2$ |
| $W$                     | real part of the Laplace variable ( $w = W + iy$ in §3, $u = W(1 + i\eta)$ in §4)                                                                                                         |
| $v; V$                  | scaled saddle: $v = NW$ in §3; $V = pnW = 5nW$ in §4                                                                                                                                      |
| $\lambda$               | $m/N$ (§4; the band is $\lambda < 4$ )                                                                                                                                                    |
| $\mathcal{L}; L$        | $\mathcal{L}(u) = 3 \sum_{k \leq 3n, 3 \nmid k} \log(1 - e^{ku})$ (cubic centre); $L$ the radial parameter $r = e^{-L/(5n)}$ (Third centre)                                               |
| $\text{Li}_s$           | polylogarithm, $\text{Li}_1(x) = -\log(1-x)$                                                                                                                                              |
| $B_1, B_2; \tilde{B}_k$ | $B_1(x) = x - \frac{1}{2}$ , $B_2(x) = x^2 - x + \frac{1}{6}$ ; $\tilde{B}_k$ their 1-periodic extensions                                                                                 |
| class                   | $m \bmod p$                                                                                                                                                                               |

TABLE 1. Notation shared by Sections 3 and 4.

**Lemma 2.2** (palindromy).  $P_{p,n}^\delta$  is palindromic of degree  $D = \delta(p-1)pn^2/2$ . The map  $m \mapsto D-m$  preserves the class 0 and maps the class  $s$  to  $-s$ . Hence the sign patterns of Conjectures 1.1 and 1.2 need only be proved for  $0 \leq m \leq D/2$ .

*Proof.*  $P_{p,n}$  is a product of  $(p-1)n$  factors  $1 - q^k$ , each satisfying  $q^k(1 - q^{-k}) = -(1 - q^k)$ , and  $\sum_{k \leq pn, p \nmid k} k = \frac{pn(pn+1)}{2} - \frac{pn(n+1)}{2} = \frac{(p-1)pn^2}{2}$ . For  $p=3$  the sign is  $(-1)^{2n \cdot 3} = 1$  and  $D = 9n^2$ ; for  $p=5$  it is  $(-1)^{4n} = 1$  and  $D = 10n^2$ . In both cases  $p \mid D$ . For  $p=3$  the classes 1, 2 are exchanged and carry the same conjectured sign; for  $p=5$  the classes  $1 \leftrightarrow 4$  and  $2 \leftrightarrow 3$  are exchanged, and again the conjectured sign is invariant under  $s \mapsto -s$ .  $\square$

**Lemma 2.3** (vanishing classes). (a) ( $p=3, \delta=3$ .)  $s(i) = [q^i]G_3^3 = 0$  for  $i \equiv 2 \pmod{3}$ .  
(b) ( $p=5, \delta=1$ .)  $s(i) = [q^i]G_5 = 0$  for  $i \equiv 3, 4 \pmod{5}$ .

*Proof.* (a) By Jacobi's identity  $(q; q)_\infty^3 = \sum_{j \geq 0} (-1)^j (2j+1) q^{j(j+1)/2}$ , and  $j(j+1)/2 \equiv 0, 1, 0 \pmod{3}$  for  $j \equiv 0, 1, 2$ . So  $(q; q)_\infty^3$  is supported on the classes  $\{0, 1\}$ , and  $G_3^3 = (q; q)_\infty^3 / (q^3; q^3)_\infty^3$  is this series times a series in  $q^3$ .

(b) By Euler's pentagonal theorem,  $(q; q)_\infty = \sum_{k \in \mathbb{Z}} (-1)^k q^{k(3k-1)/2}$ . Modulo 5 the exponent  $k(3k-1)/2$  takes the values 0, 1, 0, 2, 2 for  $k \equiv 0, 1, 2, 3, 4$ . So  $(q; q)_\infty$  is supported on the residues  $\{0, 1, 2\}$ , and multiplying by  $1/(q^5; q^5)_\infty$ , a series in  $q^5$ , preserves this.  $\square$

In the vanishing (“thin”) classes the main term of  $s(m)$  is absent and the sign of  $c_n(m)$  is produced by the convolution with  $E_n$ .

**2.3. The circle-method expansion of the eta-quotient.** Let  $f(x) = \prod_{m \geq 1} (1 - x^m)^{-1}$ ,  $\tau = h/k + iz/k^2$  with  $\operatorname{Re} z > 0$ ,  $(h, k) = 1$ , and  $d = \gcd(p, k)$ . The eta transformation, in the form of [11, Prop. 7] with the Hardy–Ramanujan convention  $hh' \equiv -1 \pmod{k}$  [9], gives

$$G_p(e^{2\pi i \tau})^\delta = \left(\frac{p}{d}\right)^{\delta/2} e^{\pi i \delta (-s(h,k) + s(ph/d, k/d))} \exp\left(\frac{\pi \delta (d^2 - p)}{12pz} - \frac{(p-1)\pi \delta z}{12k^2}\right) \widehat{G}^\delta, \quad (1)$$

where

$$\widehat{G} = \frac{f(e^{2\pi i d h'_d/k - 2\pi d^2/(pz)})}{f(e^{2\pi i h'/k - 2\pi/z})},$$

with  $s(h, k)$  the Dedekind sum and principal powers. For our two cases this has the following structure.

- *Growing cusps* ( $p \mid k$ ,  $d = p$ ). The prefactor is unimodular and the exponential is  $\exp(\pi B/z - \pi Bz/k^2)$  with  $B = \delta(p-1)/12$ . The first correction term of  $\widehat{G}^\delta$  grows only if  $\delta > 24/(p-1)$ , which is not the case for (3, 3) or (5, 1). Integrated over the Ford circle, the leading term gives the Bessel main term  $\mathcal{P}_k(i)$ , since  $\oint_K \exp(\pi B/z + \pi Cz/k^2) dz = \mathcal{P}_k(i)$  for the circle  $K = \{|z - \frac{1}{2}| = \frac{1}{2}\}$  oriented clockwise (Lemma 4.10).
- *Bounded cusps* ( $p \nmid k$ ). On  $\operatorname{Re}(1/z) \geq 1$ ,  $|G_p^\delta| \leq K_p^\delta$  with  $K_p = \sqrt{p} e^{-\pi(p-1)/(12p)} f(e^{-2\pi/p}) f(e^{-2\pi})$ .

Carrying out the Rademacher dissection with Farey order at least  $\sqrt{4\pi i} + 1$  and moving each integral to the chord of its Ford arc gives an expansion of the form

$$s(i) = \mathfrak{a}(i) \mathcal{P}_p(i) + \rho(i), \quad |\rho(i)| \leq \sum_{\substack{2p \leq k \leq K(i) \\ p \mid k}} \varphi(k) \mathcal{P}_k(i) + E_{\text{con}}, \quad (2)$$

with an explicit constant  $E_{\text{con}} = E_{\text{arc}} + E_{\text{rem}} + E_{\text{ng}}$  (arc completion, remainder at growing cusps, bounded cusps). The two instances are Theorem 3.4 ( $p = 3$ ; proved in §3.6) and Theorem 4.3 ( $p = 5$ ; proved in full in §4.2). The main-term amplitude  $\mathfrak{a}(i)$  at the cusps of denominator  $p$  is computed exactly from the Dedekind sums  $s(h, p)$ , and it vanishes precisely on the thin classes of Lemma 2.3.

**2.4. The region decomposition.** For each  $n$  the half-range  $0 \leq m \leq D/2$  is covered by the following regions (the precise thresholds are in Tables 2 and 3).

- (1) **Exact range**,  $n \leq n_0$ : exact computation of  $P_{p,n}^\delta$  in  $\mathbb{Z}[q]$ .
- (2) **Band**,  $m < cN$  with  $c = 2$  ( $p = 3$ ) or  $c = 4$  ( $p = 5$ ): since  $e(j) = 0$  for  $0 < j < N$ , only partitions of  $j$  into at most  $c-1$  parts from  $E_n$  contribute, and  $c_n(m)$  is a short explicit sum of values  $s(i)$ .
- (3) **Laplace region**: the main term of (2) is represented as a Laplace-type integral (Poisson form for  $p = 3$ , Bessel–Laplace form for  $p = 5$ ), convolved with  $E_n$  at a twisted point, and evaluated by a saddle-point analysis in scaled variables. The remainder  $\rho$  is bounded by Rankin’s trick at the same saddle.
- (4) **Centre**: Cauchy’s formula for the polynomial, evaluated through the saddles near the primitive  $p$ -th roots of unity, together with bounds for the rest of the circle.

**2.5. Ball-arithmetic certificates.** Every inequality on which an analytic region rests is reduced to finitely many inequalities between explicit functions of a few real parameters and verified in ball arithmetic (Arb [7], via `python-flint`). A ball computation returns an enclosure of the exact value over the whole parameter ball; an inequality is *certified* on a ball when the enclosure lies strictly on the correct side. The parameter domains are covered by finitely many balls (tiles);

failing balls are bisected. Every program aborts on a failed assertion and prints a summary line, and it is that line which constitutes the certificate (§5).

Uniformity in  $n$  uses two devices.

- *Analyticity in  $1/N$ .* After scaling, the integrands are analytic in  $\epsilon = 1/N$  at  $\epsilon = 0$ , so one ball  $\epsilon \in [0, \epsilon_1]$  covers all  $N \geq 1/\epsilon_1$  (§3).
- *Monotonicity.* Every term depending on  $n$  (or on a scaled size parameter  $X$ ) is of the form  $n^a e^{-nb}$ , or a finite sum of such terms, and a certified inequality on the exponents shows that it is non-increasing in  $n$  beyond the base point. One evaluation at the base point then covers all larger  $n$ .

### 3. THE CUBIC CASE: THE CLASS $m \equiv 2 \pmod{3}$

In this section  $p = \delta = 3$ ,  $P_n = P_{3,n}$ ,  $c_n(m) = [q^m]P_n^3$ ,  $N = 3n + 1$ ,  $a_3 = \pi^2/9$ ,  $\omega = \omega_3 = e^{2\pi i/3}$ ,  $s(i) = [q^i]G_3^3$  and  $P_\infty = G_3$ , so that  $E_n = P_\infty^3/P_n^3 = \prod_{k>3n, 3 \nmid k} (1 - q^k)^{-3}$ . By Lemma 2.3(a),  $c_n(m)$  for  $m \equiv 2$  is produced entirely by the tail  $E_n$ .

**Theorem 3.1.** *For all  $n \geq 1$  and all  $m \equiv 2 \pmod{3}$  with  $m \leq \frac{9}{2}n^2$  we have  $c_n(m) = 0$  if  $m < 3n$  and  $c_n(m) < 0$  if  $m \geq 3n$ . Together with [14] this proves Conjecture 1.1.*

Put  $\bar{\mu} = m/N^2$  and  $\tilde{\mu} = (m - N - \frac{1}{4})/N^2$ . The admissible pairs  $(n, m)$  are covered by five regions (Table 2).

| region                                                                         | method                              | where      |
|--------------------------------------------------------------------------------|-------------------------------------|------------|
| $n \leq 243$                                                                   | exact integer computation           | §3.2       |
| $n \geq 244, m < 6n + 2$                                                       | band lemma                          | Prop. 3.10 |
| $n \geq 244, m \geq 6n + 2, \tilde{\mu} < 0.0985, \text{ saddle } v \leq 3000$ | Laplace certificate, uniform in $n$ | §3.3       |
| $n \geq 244, \text{ saddle } v \geq 3000$                                      | closing lemma                       | §3.3.3     |
| $n \geq 244, \bar{\mu} \geq 0.098$                                             | centre certificate, uniform in $n$  | §3.4       |

TABLE 2. The covering of the admissible  $(n, m)$  for  $p = 3$ . That these regions exhaust all cases is Proposition 3.18.

#### 3.1. Structural lemmas.

**Lemma 3.2** (tail phase). *For  $0 < q < 1$ ,  $\Delta(q) := \text{Im} \log E_n(\omega q) \in (0, \pi/2)$ .*

*Proof.*  $\text{Im} \log E_n(\omega q) = 3 \sum_{A \geq n} [f(q^{3A+1}) - f(q^{3A+2})]$  with  $f(x) = \arctan(\sqrt{3}x/(2+x))$ , increasing on  $[0, 1]$  with  $f(1) = \pi/6$ . Each bracket is positive, and  $f(q^{3A+1}) - f(q^{3A+2}) \leq f(q^{3A+1}) - f(q^{3A+4})$  telescopes to at most  $f(q^{3n+1}) < \pi/6$ .  $\square$

**Lemma 3.3** (closed forms). *Let  $\chi'(r) = 1$  for  $r \equiv 1$ ,  $-1$  for  $r \equiv 2$ , and  $h_r(q) = (q^{Nr} + q^{(N+1)r})/(1 - q^{3r})$ . For  $|q| < 1$ ,*

$$\log E_n(\omega q) = S(q) + i\Delta(q), \quad S = - \sum_{3 \nmid r} \frac{3}{2r} h_r + \sum_{3 \mid r} \frac{3}{r} h_r, \quad \Delta = \sum_{3 \nmid r} \frac{3\sqrt{3}}{2r} \chi'(r) \frac{q^{Nr}(1 - q^r)}{1 - q^{3r}},$$

*as an identity of analytic functions. Consequently  $F(q) := \frac{1}{2i}(E_n(\omega q) - E_n(\bar{\omega} q)) = e^S \sin \Delta$  on  $0 < q < 1$ , and  $F$  is the analytic continuation of this expression.*

*Proof.* Expand  $-3 \log(1 - \omega^k q^k) = 3 \sum_r \omega^{kr} q^{kr}/r$  and sum the geometric series over  $k \equiv 1, 2 \pmod{3}$ ,  $k > 3n$ ; this gives  $\sum_r \frac{3}{r} (\omega^r q^{Nr} + \omega^{2r} q^{(N+1)r})/(1 - q^{3r})$ , whose real and imaginary parts for real  $q$  are the displayed expressions, and the displayed combination equals the series identically in  $q$ .  $\square$

Let  $\chi(i) = 1, -1, 0$  for  $i \equiv 0, 1, 2 \pmod{3}$ . With  $B = \frac{1}{2}$  and  $C = 2i - \frac{1}{2}$  we have  $\mathcal{P}_3(i) = \sqrt{a_3/(i - \frac{1}{4})} I_1(2\sqrt{a_3(i - \frac{1}{4})})$ .

**Theorem 3.4** (explicit expansion of  $G_3^3$ ). *For all  $i \geq 1$ ,*

$$s(i) = \sqrt{3}\chi(i)\mathcal{P}_3(i) + \rho(i), \quad |\rho(i)| \leq \sum_{\substack{6 \leq k \leq K(i) \\ 3|k}} \varphi(k)\mathcal{P}_k(i) + E_{\text{con}},$$

with  $K(i) = \lfloor \sqrt{4\pi i} \rfloor + 2$  and  $E_{\text{con}} = 50.1501$ .

The proof is in §3.6. Thus  $\mathfrak{a}(i) = \sqrt{3}\chi(i)$  in the notation of (2).

**Lemma 3.5** (Poisson form). *Let  $\mathcal{P}^\sharp(q) = \sum_{i \geq 1} \mathcal{P}_3(i)q^i$ . For  $\text{Re } w > 0$ ,  $|\text{Im } w| \leq \pi$ ,*

$$\mathcal{P}^\sharp(e^{-w}) = e^{-w/4}(e^{a_3/w} - 1 + \varepsilon(w)), \quad \varepsilon(w) = \sum_{k \neq 0} (-i)^k (e^{a_3/(w+2\pi ik)} - 1),$$

and  $|-1 + \varepsilon(w)| \leq 1.42$ .

*Proof.*  $x \mapsto \sqrt{a_3/x} I_1(2\sqrt{a_3x})$  has Laplace transform  $e^{a_3/w} - 1$ ; Poisson summation over the shifted lattice  $i - \frac{1}{4}$  gives the identity. The bound is Certificate 3.6.  $\square$

**Certificate 3.6.** Ball-arithmetic enclosure of  $|-1 + \varepsilon(w)|$  on 1600 boxes covering  $0 < \text{Re } w \leq 10$ ,  $|\text{Im } w| \leq \pi$ , with the terms  $|k| > 400$  bounded by  $8a_3/(\pi \cdot 400) + a_3^2 e^{a_3/\pi}/(400\pi^2)$ . Output:  $|-1 + \varepsilon| \leq 1.420$ .

*Certificate:* `poisson_check.py`; `log_cubic/logs/04a_poisson.log`.

**Proposition 3.7** (reduction). *Let  $m \equiv 2 \pmod{3}$ ,  $W > 0$  and  $w = W + iy$ . Then  $c_n(m) = -2\text{Im } J + R(m)$ , where*

$$\text{Im } J = \frac{1}{2\pi} \int_{-\pi}^{\pi} \text{Re } e^{\widehat{R}(w)} dy + \text{Im } J_\varepsilon, \quad \widehat{R}(w) = \frac{a_3}{w} + \left(m - \frac{1}{4}\right)w + S(e^{-w}) + \log \sin \Delta(e^{-w}),$$

$|\text{Im } J_\varepsilon| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} 1.42 e^{(m-\frac{1}{4})W} |F(e^{-w})| dy$ , and, for every  $w_1 > 0$ ,

$$|R(m)| \leq \sum_{\substack{6 \leq k \leq K(m) \\ 3|k}} 3\varphi(k) e^{mw_1 + a_k/w_1} E_n(e^{-w_1}) + E_{\text{con}} e^{mw_1} E_n(e^{-w_1}), \quad a_k = \pi^2/k^2.$$

*Proof.*  $c_n(m) = \sum_j e(j)s(m-j)$  with  $e(j) = [q^j]E_n \geq 0$ . Inserting Theorem 3.4, the  $\chi$ -part equals  $\sum_j e(j)\sqrt{3}\chi(m-j)\mathcal{P}_3(m-j) = -2\text{Im}[q^m]\mathcal{P}^\sharp(q)E_n(\omega q)$  for  $m \equiv 2$  (compare coefficients:  $e(j)$  is real and  $\sqrt{3}\chi(m-j) = -2\text{Im } \omega^j$  for  $m \equiv 2$ ). The coefficient is  $\frac{1}{2\pi} \int \mathcal{P}^\sharp(e^{-w})E_n(\omega e^{-w})e^{mw} dy$ ; by Lemma 3.5, reality of  $[q^m]\mathcal{P}^\sharp F$  and Lemma 3.3 this gives the displayed  $\text{Im } J$ . For  $R$ , the remainder terms are bounded termwise using  $e(j) \geq 0$ ,  $\mathcal{P}_k \geq 0$  and  $\sum_i \mathcal{P}_k(i)e^{-w_1 i} \leq (1+2a_k)e^{a_k/w_1} \leq 3e^{a_k/w_1}$  (from  $I_1(y) \leq \frac{y}{2}e^y$  and the Laplace transform).  $\square$

**Remark 3.8.** Proposition 3.7 is checked end to end by `identity_check.py`: for  $n = 12$  ( $m = 200, 302, 404$ ) and  $n = 20$  ( $m = 500, 800, 1100$ ) the exact  $c_n(m)$  agrees with  $-2(\text{Im } J_0 + \text{Im } J_\varepsilon) + R(m)$ , each piece computed independently, to relative error  $< 10^{-8}$ . The bound of Theorem 3.4 is checked exactly for all  $i \leq 3000$  and for 3388 values  $i \leq 108000$  (`expansion_check.py`); the maximal ratio is  $\sqrt{3}/2$ . Neither check is part of the proof.

In particular,  $c_n(m) < 0$  follows from

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \text{Re } e^{\widehat{R}(W+iy)} dy > |\text{Im } J_\varepsilon| + \frac{1}{2}|R(m)|. \quad (3)$$

### 3.2. The finite range and the band.

**Certificate 3.9** (finite range).  $P_n^3$  is computed exactly in  $\mathbb{Z}[q]$  (FLINT) for  $n \leq 243$ , and it is checked that  $c_n(m) = 0$  for  $m < 3n$  and  $c_n(m) < 0$  for  $3n \leq m \leq \frac{9}{2}n^2$ ,  $m \equiv 2$ . Output: no violations.

*Certificate:* `exact_small_n.py; log cubic/logs/01_exact_small_n_{a,b,c}.log`.

**Proposition 3.10** (band). For  $3n + 2 \leq m < 6n + 2$  with  $m = 3M + 2$ ,  $c_n(m) = 3 \sum_{i=0}^{M-n} g(i)$  with  $g(i) = s(3i) + s(3i + 1)$ , and  $g(i) < 0$  for all  $i \geq 0$ . Hence  $c_n(m) < 0$  in the band.

*Proof.* Since  $6n + 2 = 2N$ ,  $E_n \equiv 1 + 3 \sum_{N \leq j < 2N, 3 \nmid j} q^j \pmod{q^{2N}}$ , and  $s(m) = 0$ . Grouping  $j = 3A + 1$  and  $j = 3A + 2$  gives the identity (checked exactly for  $n \leq 60$  by `band_lemma.py`). Negativity of  $g$  is exact for  $i \leq 36000$  (Certificate 3.11). For  $i \geq 36000$ , Theorem 3.4 gives  $g(i) \leq -\sqrt{3}(\mathcal{P}_3(3i + 1) - \mathcal{P}_3(3i)) + \rho_{\max}(3i) + \rho_{\max}(3i + 1)$ . Since  $\mathcal{P}_3(x) = 2a_3 I_1(y)/y$  with  $y = 2\sqrt{a_3(x - \frac{1}{4})}$ , we have  $\mathcal{P}_3'(x) = 4a_3^2 I_2(y)/y^2$ , and  $\mathcal{P}_3$  is increasing and convex, so the difference is at least  $\mathcal{P}_3'(3i)$ . Using  $I_2(y) \geq \kappa_2 e^y / \sqrt{2\pi y}$  for  $y \geq y_0$ ,  $I_1(y) \leq e^y / \sqrt{2\pi y}$ ,  $\mathcal{P}_k \leq \mathcal{P}_6$  for  $k \geq 6$  and  $\sum \varphi(k) \leq K^2/2$ , the ratio of the  $\rho$ -bound to the main difference is of the form  $\text{poly}(y)e^{-y/2}$ , decreasing for  $y \geq 9$ , and at  $i = 36000$  ( $y_0 = 688.3$ ) it is at most  $5 \cdot 10^{-139}$  in ball arithmetic.

*Certificate:* `band_tail_rigorous.py; log cubic/logs/02b_band_tail.log`. □

**Certificate 3.11.** Exact  $s(i)$  from Jacobi's identity and the recurrence  $jb(j) = 3 \sum_t \sigma(t)b(j - t)$  for  $b = [q^j](q; q)_\infty^{-3}$ . Output: the identity of Proposition 3.10 holds for  $n \leq 60$ ;  $g(i) < 0$  for all  $i \leq 36000$ .

*Certificate:* `band_lemma.py 36000; log cubic/logs/02_band_lemma.log`.

### 3.3. The Laplace regime.

3.3.1. *Scaled form.* Let  $\zeta = Nw$ ,  $\epsilon = 1/N$ ,  $g(x) = x/(1 - e^{-x})$  and  $h(x) = (1 - e^{-x})/(1 - e^{-3x}) = g(3x)/(3g(x))$ . Both  $g$  and  $h$  are analytic for  $|x| < 2\pi/3$ .

**Lemma 3.12.**  $S(e^{-w}) = N\tilde{S}(\zeta, \epsilon)$  and  $\hat{R}(w) = N\mathcal{G}(\zeta) + (m - N - \frac{1}{4})\zeta/N$ , where

$$\tilde{S} = \sum_r s_r e^{-r\zeta} (1 + e^{-r\zeta\epsilon}) \frac{g(3r\zeta\epsilon)}{3r\zeta}, \quad \tilde{Q} = \sum_{3 \nmid r} \frac{3\sqrt{3}}{2r} \chi'(r) e^{-(r-1)\zeta} h(r\zeta\epsilon),$$

$s_r = -\frac{3}{2r} (3 \nmid r)$ ,  $\frac{3}{r} (3 \mid r)$ , and  $\mathcal{G} = \frac{a_3}{\zeta} + \tilde{S} + \epsilon(\log \tilde{Q} + \log \frac{\sin \Delta}{\Delta})$  with  $\Delta = \tilde{Q}e^{-\zeta}$ . The saddle condition  $\hat{R}'(W) = 0$  reads  $-\mathcal{G}'(v) = \tilde{\mu}$  with  $v = NW$ .

*Proof.* Substitute  $q = e^{-\zeta\epsilon}$  in Lemma 3.3;  $\log \sin \Delta = \log \tilde{Q} - \zeta + \log \frac{\sin \Delta}{\Delta}$ , and the  $-\zeta$  combines with  $(m - \frac{1}{4})w$  to  $(m - N - \frac{1}{4})\zeta/N$ . □

Thus a pair  $(N, m)$  in this regime is represented by its real saddle  $v$  and  $\epsilon = 1/N$ . Since  $m \geq 2N$ ,  $\tilde{\mu} \geq \epsilon(1 - \frac{1}{4N}) \geq 0.999\epsilon$ ; balls with  $-\mathcal{G}'(v) < 0.999\epsilon$  on the whole ball contain no pairs.

3.3.2. *The error budget on a ball.* Fix a ball  $V_b \ni v$  and a ball  $E \ni \epsilon$  with  $E \subset [0, 1/733]$ , and let  $N_b = \max(733, 1/\sup E)$ . Write  $\mathcal{G}_j = \mathcal{G}^{(j)}(v)$ ,  $\sigma_y = N^{-3/2} \mathcal{G}_2^{-1/2}$  and  $\tau = y/\sigma_y$ . Dividing (3) by  $e^{\hat{R}(W)} \sigma_y / (2\pi)$ , it suffices to show

$$B_{\text{tot}} := \frac{Z_1 + Z_2 + Z_3 + Z_\epsilon + Z_R}{L_w} < 1, \quad (4)$$

where  $L_w$  bounds the window integral from below and the  $Z$ 's bound the remaining contributions from above.

*Window*  $|\tau| \leq c$ .  $\widehat{R}(W + i\sigma_y\tau) - \widehat{R}(W) = -\frac{\tau^2}{2} - i\mathcal{G}_{3,n}\frac{\tau^3}{6} + \phi_4\frac{\tau^4}{24} + E_5$  with  $\mathcal{G}_{3,n} = \sqrt{\epsilon}|\mathcal{G}_3|\mathcal{G}_2^{-3/2}$ ,  $\phi_4 = \epsilon\mathcal{G}_4\mathcal{G}_2^{-2}$ ,  $|E_5| \leq \mathcal{G}_{5,n}|\tau|^5/120$ ,  $\mathcal{G}_{5,n} = \epsilon^{3/2}M_5\mathcal{G}_2^{-5/2}$ , where  $M_5$  bounds  $|\mathcal{G}^{(5)}|$  on the window:  $120a_3/v^6$  for the  $a_3/\zeta$  part plus a Cauchy estimate (Lemma B.9) for the rest on the circle  $|\zeta - v| = 0.6v$ .  $L_w = \int_{-c}^c \ell(\tau)d\tau$  with  $\ell$  the resulting pointwise lower bound of the real part ( $\cos \theta \geq 1 - \theta^2/2$ ).

*Zone 1*,  $c \leq |\tau|$ ,  $|t| \leq t_s$  ( $t = Ny$ ).  $\text{Re}[\mathcal{G}(v + it) - \mathcal{G}(v)] \leq -\frac{t^2}{2}(\kappa_0 - \epsilon\kappa_b)$  from the even Taylor terms of  $\mathcal{G}_0 = a_3/\zeta + \widetilde{S}$  and of  $b = \log \widetilde{Q} + \log \frac{\sin \Delta}{\Delta}$  with Cauchy remainders at order 6; thus  $Z_1 = 2 \int_c^\infty e^{-\kappa\tau^2/2}d\tau$ ,  $\kappa = (\kappa_0 - \epsilon\kappa_b)/\mathcal{G}_2$ , independent of  $N$ .

*Zone 2*,  $t_s \leq t \leq t_C$ . On cells,  $\text{Re} \widehat{R} - \widehat{R}(W) = NA(t) + B(t)$  with  $A = \text{Re}[\mathcal{G}_0(v + it) - \mathcal{G}_0(v)]$  and  $B = \text{Re}[b(v + it) - b(v)]$ , enclosed in ball arithmetic (the  $a_3/\zeta$  part in closed form, the factor  $e^{-rv}$  separated to control dependency). The certificate asserts  $A < -1/(2N_b)$  on every cell, so  $N^{1/2}e^{NA} \leq N_b^{1/2}e^{N_bA}$  for  $N \geq N_b$ , and  $Z_2 = (N_b\mathcal{G}_2)^{1/2} \sum_{\text{cells}} 2(e^{N_bA+B} + 3e^{N_b(A-\text{Re}(a_3/\zeta))+B})|\text{cell}|$  (the second term is the  $\varepsilon$ -part).

*Zone 3*,  $t \geq t_C$ . Either (crude)  $|F| \leq E_n(e^{-W})$  and  $|e^{a_3/w} - 1 + \varepsilon| \leq e^{a_3W/(W^2+y^2)} + 3$ , giving an exponent  $N\beta + O(1)$  with  $\beta = \Psi(v)/v + a_3v/(v^2 + t_C^2) - a_3/v - \widetilde{S}(v)$ ,  $\Psi(v) = 2e^{-v}/(1 - e^{-v})$ , asserted  $\beta < -3/(2N_b)$ ; or (split, when  $\inf E > 0$ )  $|\widetilde{Q}| \leq \sum_{3|r} \frac{3\sqrt{3}}{2r} e^{-(r-1)v} 2(1/(3rv \inf E) + 1)$  and  $|\widetilde{S}|$  bounded termwise, with slope  $< -5/(2N_b)$ .

$R(m)$ . Proposition 3.7 with  $w_1 = t'W$ ,  $t'$  from a fixed list,  $\log E_n(e^{-w_1}) \leq N\Psi(Nw_1)/(Nw_1) + 6|\log(1 - e^{-Nw_1})|$ ,  $\tilde{\mu} = -\mathcal{G}'(v)$  and  $\varphi$ -sum  $\leq 3(K^2/6 + K)$ ; each exponent is asserted to have slope  $< -7/(2N_b)$ .

**Lemma 3.13.** *If all assertions above hold on  $(V_b, E)$ , then  $B_{\text{tot}}$  computed with  $N = N_b$  bounds (4) for every  $N$  with  $1/N \in E$  and every  $m$  whose saddle lies in  $V_b$ .*

*Proof.*  $L_w$  and  $Z_1$  do not depend on  $N$  (their  $N$ -dependence enters only through  $\epsilon \in E$ ). Every other term is of the form  $N^p e^{N\lambda}$  with  $\lambda$  bounded by the asserted slope  $< -p/N_b$ , hence non-increasing in  $N \geq N_b$ .  $\square$

**Certificate 3.14** (uniform Laplace grid).  $v \in [2.95, 3000]$  is covered by 36 chunks of balls and, for each  $v$ -ball,  $\epsilon \in [0, 1/733]$  by  $\epsilon$ -balls; balls whose pairs are vacuous are discarded only after the rigorous bound  $\tilde{\mu} \leq a_3/v^2 + \max_{|\zeta-v|=v/2} |h(\zeta) - h(v)|/(v/2)$  (with  $h = \widetilde{S} + \epsilon b$ ) proves  $\tilde{\mu} < 0.999\epsilon$ , or when  $\tilde{\mu} > 0.0985$  (centre regime). Failing balls are bisected. Output: every chunk certified, 0 unresolved balls,  $\max B_{\text{tot}} = 0.117$ .

*Certificate:* `run.uniform.py; log cubic/logs/uni*.log` (36 logs).

### 3.3.3. The closing lemma.

**Proposition 3.15.** *For every pair with saddle  $v \geq 3000$ , (4) holds with  $B_{\text{tot}} \leq 0.0082$ .*

*Proof and certificate.* Put  $s = \epsilon v^2/a_3$ ; the constraint  $\tilde{\mu} \geq 0.999\epsilon$  and  $\tilde{\mu} = (a_3/v^2)(1 + O(e^{-v}))$  give  $s \leq 1.0011$ . On  $|\zeta - v| \leq v/2$ :  $|\widetilde{S}| \leq \sum_r \frac{3}{r} e^{-rv/2} 2(\frac{2}{3rv} + \epsilon) \leq 5 \cdot 10^{-655}$ ;  $|h(x)| \leq 4$  for every  $r$  (by the Bernoulli enclosure if  $|x| < 0.9$ , and  $|h(x)| \leq 2(|x|/(3\text{Re } x) + 1)$  with  $|x|/\text{Re } x = |\zeta|/\text{Re } \zeta \leq 3$  otherwise), so the  $r \geq 2$  part of  $\widetilde{Q}$  and  $\Delta$  are  $\leq 4 \cdot 10^{-651}$ ; and  $\log h$  on the disc  $|x| \leq \frac{3}{2}a_3s/v$  is enclosed, giving  $|b(\zeta) - b(v)| \leq 4.4 \cdot 10^{-3}$ . By Lemma B.9, the relative perturbations of  $\mathcal{G}_2, \dots, \mathcal{G}_5$  from their pure  $a_3/\zeta$  values are  $\leq 4.7 \cdot 10^{-5}$ , so  $\mathcal{G}_{3,n} \leq 0.0388$ ,  $\phi_4 \leq 2.0 \cdot 10^{-3}$ ,  $\mathcal{G}_{5,n} \leq 1.03 \cdot 10^{-3}$ ,  $\kappa \geq 0.8(1 - 10^{-5})$  on  $|t| \leq v/2$ , and  $L_w \geq 2.494$ ,  $Z_1 \leq 0.0205$ . For  $|t| \geq v/2$  the exponent is at most  $-\frac{v}{5s}(1 - 10^{-5}) + \log(\frac{4v}{3a_3s} + 6) + 20ve^{-v}/(a_3s)$  with prefactor  $\sqrt{2}v^{3/2}/(a_3s^{3/2})$ ; its  $s$ -derivative is positive for  $s \leq 1.0011$  because  $v/5 > 2.5 \cdot 1.0011 + 20ve^{-v}/a_3$ , and it decreases in  $v$ , so the worst case is  $v = 3000$ ,  $s = 1.0011$ , where  $Z_2 \leq 5 \cdot 10^{-251}$ . The  $R$ -term is  $\leq 10^{-1278}$ . All numbers are ball enclosures.

*Certificate:* ray\_asymptotic2.py; log\_cubic/logs/07\_closing\_lemma.log.

□

**3.4. The centre regime.** Here  $c_n(m) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_n(re^{i\theta})^3 r^{-m} e^{-im\theta} d\theta$  is treated directly, without Proposition 3.7. Put  $\mathcal{L}(u) = 3 \sum_{k \leq 3n, 3 \nmid k} \log(1 - e^{ku})$ ,  $u_0 = 2\pi i/3$ , and let  $u^* = u_0 + x^*$  be the saddle  $\mathcal{L}'(u^*) = m$  near  $u_0$ ; write  $\xi = -N \operatorname{Re} x^*$  and  $\eta = N \operatorname{Im} x^*$ . The conjugate saddle near  $-u_0$  contributes the complex conjugate.

**Lemma 3.16** (sign margin). *The phase of the main term is  $\alpha(r) - \frac{2\pi}{3} + b + E_1 \pmod{2\pi}$ , where  $\alpha(r) = -\frac{\pi}{6} + \arctan(\sqrt{3}r^{3n}/(2+r^{3n}))$ ,  $b = -\frac{1}{2} \arg \mathcal{L}''(u^*)$ , and  $|E_1| \leq \sup |\mathcal{L}''| \delta^2/2$  with  $\delta = \operatorname{Im} x^*$ ; moreover  $|\delta| \leq 3N \operatorname{TV}_s / (N^3 \rho_{\text{lo}} - 3N^2 V_K)$  by the saddle equation and Abel summation.*

*Proof.* The main term is  $M = e^{\mathcal{L}(u^*) - mu^*} \omega e^{ib} \sqrt{2\pi/|\mathcal{L}''(u^*)|} / (2\pi)$  (steepest descent through  $u^*$  in the direction  $ie^{ib}$ ). Write  $u'_0 = \log r + 2\pi i/3$ , so  $u^* = u'_0 + i\delta$ . Then  $\operatorname{Im}[\mathcal{L}(u^*) - mu^*] = \operatorname{Im} \mathcal{L}(u'_0) + \int_0^\delta (\operatorname{Re} \mathcal{L}'(u'_0 + is) - m) ds - m \cdot \frac{2\pi}{3}$ , and  $\operatorname{Re} \mathcal{L}'(u^*) = m$  gives  $|\operatorname{Re} \mathcal{L}'(u'_0 + is) - m| \leq \sup |\mathcal{L}''| |\delta - s|$ , whence the bound on  $E_1$ . Next  $\operatorname{Im} \mathcal{L}(u'_0) = \arg P_n(r\omega)^3 = \alpha(r)$  by the pairing  $(1 - r^{3j+1}\omega)(1 - r^{3j+2}\omega^2)$  of Lemma 3.2 type;  $m \equiv 2$  gives  $m \cdot \frac{2\pi}{3} \equiv \frac{4\pi}{3}$ , and  $\arg \omega = \frac{2\pi}{3}$ . For  $\delta$ :  $\operatorname{Im} \mathcal{L}'(u'_0) = 3 \sum_i [(3i+2)s_\omega(x_{3i+2}) - (3i+1)s_\omega(x_{3i+1})]$  with  $s_\omega(x) = \operatorname{Im}(\omega x/(1 - \omega x))$  and  $x_k = r^k$ , which is at most  $3N \operatorname{TV}_\tau(\tau s_\omega(e^{-\xi\tau}))$  in modulus; and  $\operatorname{Im} \mathcal{L}'(u'_0 + i\delta) = 0$  with  $\operatorname{Re} \mathcal{L}'' \geq N^3 \rho_{\text{lo}} - 3N^2 V_K$  on the segment. □

(Here  $\rho_{\text{lo}}$  is a lower bound for the scaled continuum curvature and  $V_K$  a Koksma total-variation constant, Lemma B.11.)

The certificate splits the circle into a window  $|\theta - \theta^*| \leq c\sigma$ , a middle stretch up to  $|\theta - 2\pi/3| \leq 0.05$ , and the far region. Each contribution is bounded uniformly in  $N \geq 733$  as a function of  $\xi$ :

- *Window:* the cubic Taylor constant  $G_3 = H_3 \sigma^3$  with  $H_3(\text{window}) \leq N^4[h_3 + 3(\operatorname{TV} + S)/N]$  and  $\sigma^3 \leq (N^3 a_2 - 3N^2 V_K)^{-3/2}$  (continuum integrals plus Koksma remainders, Lemma B.11); the window shrinks with  $N$ .
- *Middle:* three zones in  $t = N(\theta - 2\pi/3)$ : Taylor with  $\operatorname{Re} \mathcal{L}'' \geq N^3 \rho_{\text{lo}}(t) - 3N^2 V_K$  ( $|t| \leq 0.8$ ); the linearised continuum profile  $-N \int_0^1 g(e^{-\xi\tau})(1 - \cos \tau t) d\tau$  plus a Koksma constant ( $0.8 \leq t \leq 10$ , slope asserted); a Dirichlet-kernel bound ( $t \geq 10$ , slope asserted).
- *Far:* for  $\theta \geq 0.05$  a Dirichlet-kernel bound; near  $\theta = 0$  a continuum profile  $-N \int g(1 + 2 \cos \tau t)$  with positivity asserted on  $t \in [0, 40]$  and a kernel bound beyond; slopes asserted.
- *Sign margin:* Lemma 3.16 with  $r^{3n} = e^{-\xi(1-1/N)}$  enclosed over all  $N \geq 733$ .

**Certificate 3.17** (centre tables). On 74  $\xi$ -balls (with finer sub-tables of 115 balls for the middle and window constants) covering  $[-0.3, 3.4]$ , the programs produce uniform bounds for the middle stretch, the window constant, the sign margin (with  $\cos < 0$  asserted) and the far region. `coverage.py` (check E) combines them into  $\text{err} = (2(\text{inner} + \text{middle}) + \text{far}) / (2|\cos|)$  on every ball. Output:  $\text{max err} = 0.49 < 1$ . The saddle offset satisfies  $|\eta| \leq 0.0155$ , within the shift margins of the tables (check G).

*Certificate:* centre\_mid\_uniform{,2,3}.py, centre\_g3\_uniform{,2,3}.py, centre\_clow\_uniform.py, coverage.py; log\_cubic/tables/\*\_table.txt, cubic/logs/08\_coverage.log.

### 3.5. Coverage and proof of Theorem 3.1.

**Proposition 3.18.** *Every  $(n, m)$  with  $n \geq 244$ ,  $m \equiv 2 \pmod{3}$ ,  $6n + 2 \leq m \leq \frac{9}{2}n^2$  lies in the Laplace regime with saddle in  $[2.95, 3000]$  or  $[3000, \infty)$ , or in the centre regime with  $\xi \in [-0.3, 3.4]$ .*

*Proof.* (A) The chunks of Certificate 3.14 tile  $[2.95, 3000]$  with no unresolved balls. (B)  $-G'(2.95) \geq 0.1015 > 0.0985$  for all  $\epsilon \in [0, 1/733]$  and  $G'' > 0$ , so every pair with  $\tilde{\mu} < 0.0985$

has  $v > 2.95$ . (C)  $m \geq 2N$  gives  $\tilde{\mu} \geq 0.999\epsilon$ . (D)  $\tilde{\mu} \geq 0.0985$  implies  $\bar{\mu} > \tilde{\mu} \geq 0.098$ . (F)  $\bar{\mu}_N(\xi) = N^{-2} \operatorname{Re} \mathcal{L}'(u^*)$  is decreasing in  $\xi$  and, by Lemma B.11,  $\bar{\mu}_N(3.4) \leq 0.0886 < 0.098$  and  $\bar{\mu}_N(-0.3) \geq 0.531 > \frac{1}{2}$  for all  $N \geq 733$ , while  $m \leq \frac{9}{2}n^2$  gives  $\bar{\mu} < \frac{1}{2}$ . Each item is checked by `coverage.py`.

*Certificate:* `coverage.py`; `log cubic/logs/08_coverage.log`.

□

*Proof of Theorem 3.1.* Certificate 3.9 for  $n \leq 243$ ; for  $n \geq 244$  Proposition 3.10 for  $m < 6n + 2$ , and otherwise Proposition 3.18 with Lemma 3.13, Certificate 3.14, Proposition 3.15 and Certificate 3.17. □

*Load-bearing hand arguments.* The proof of Theorem 3.1 depends on the following written arguments in addition to the certificates: Lemmas 3.2–3.5, Proposition 3.7, Theorem 3.4 (§3.6), Lemma 3.12, the derivation of the budget (4), Lemma 3.13, the analytic parts of Proposition 3.15, Lemma 3.16, and the elementary lemmas of Appendix B.2.

**3.6. Proof of Theorem 3.4.** We specialise (1) to  $p = \delta = 3$ ; then  $G_3^3 = P_\infty^3$ . Let  $\tau = h/k + iz/k^2$ ,  $d = \gcd(3, k)$ . The eta transformation gives

$$G_3(e^{2\pi i \tau})^3 = \left(\frac{3}{d}\right)^{3/2} e^{3\pi i(-s(h,k)+s(3h/d,k/d))} \exp\left(\frac{\pi(d^2-3)}{12z} - \frac{\pi z}{2k^2}\right) \widehat{G}^3, \quad \widehat{G} = \frac{f(e^{2\pi i d h'_d/k - 2\pi d^2/(3z)})}{f(e^{2\pi i h'/k - 2\pi/z})},$$

with  $hh' \equiv -1 \pmod{k}$  and principal powers.

*Growing and bounded cusps.* If  $3 \mid k$  then  $d = 3$ , the prefactor has modulus 1 and the exponential is  $e^{\pi/(2z)}$ : every such cusp grows with  $B = \frac{1}{2}$ , and the correction term of  $\widehat{G}^3$  grows only if  $\delta > 24/(p-1) = 12$ , which is not the case. If  $3 \nmid k$ , then on  $\operatorname{Re}(1/z) = w \geq 1$ ,  $|G_3^3| \leq K_3^3$  with  $K_3 = \sqrt{3} e^{-\pi/18} f(e^{-2\pi/3}) f(e^{-2\pi}) = 1.691256$ , the maximum being attained at  $w = 1$ .

*Rademacher dissection.* Take Farey order  $N' \geq \sqrt{4\pi i} + 1$ . On the Ford-circle arcs mapped to  $K = \{z - \frac{1}{2} \mid |z| = \frac{1}{2}\}$ , the endpoints satisfy  $|z'|, |z''| \leq \sqrt{2}k/(N' + 1)$  and  $\operatorname{Re} z \leq 2k^2/(N' + 1)^2$ , so  $|e^{\pi C z/k^2}| \leq e$  with  $C = 2i - \frac{1}{2}$ .

*Main terms.* For  $3 \mid k$ , the term 1 of  $\widehat{G}^3$  gives  $\exp(\pi/(2z) + \pi C z/k^2)$ ; completing the arc to  $K$  and using  $\frac{i}{k^2} \int_K \exp(\pi B/z + \pi C z/k^2) dz = \mathcal{P}_k(i)$  (Lemma 4.10) yields the main term with amplitude of modulus 1, for each of the  $\varphi(k)$  values of  $h$ . The completion costs at most  $\frac{1}{k^2} \cdot 2 \cdot \frac{\pi}{2} \cdot \frac{\sqrt{2}k}{N'+1} e^{1+\pi/2}$  per cusp, and  $\sum_{3|k \leq N'} \varphi(k)/k \leq \frac{2}{9}(N' + 1)$  because  $\varphi(3j)/(3j) \leq \frac{2}{3}$ ; total  $E_{\text{arc}} = \sqrt{2}\pi \frac{2}{9} e^{1+\pi/2} = 12.9103$ .

*Remainder at growing cusps.* Moving to the chord (length  $\leq 2\sqrt{2}k/(N' + 1)$ ),  $|e^{\pi/(2z)}(\widehat{G}^3 - 1)| \leq e^{\pi w/2} (f(e^{-2\pi w})^3 f(e^{-6\pi w})^3 - 1)$ , whose terms  $e^{\pi w/2} e^{-2\pi j w}$  ( $j \geq 1$ ) are non-increasing, so the maximum is at  $w = 1$ ; total  $E_{\text{rem}} = 2\sqrt{2}\frac{2}{9} e^{1+\pi/2} (f(t)^3 f(t^3)^3 - 1) = 0.0463$ ,  $t = e^{-2\pi}$ .

*Bounded cusps.* On the chords, the integrand is at most  $eK_3^3$ ; summing  $\frac{1}{k^2} \frac{2\sqrt{2}k}{N'+1}$  over  $\leq \varphi(k) \leq k$  values of  $h$  and  $k \leq N'$  gives  $E_{\text{ng}} = 2\sqrt{2}eK_3^3 = 37.1935$ .

*The  $k = 3$  amplitude.* For  $k = 3$  we have  $d = 3$ ,  $(3/d)^{3/2} = 1$  and  $s(3h/d, k/d) = s(h, 1) = 0$ , so the phase of the cusp  $h/3$  is  $e^{-3\pi i s(h,3)}$ . The Dedekind sums are  $s(1, 3) = \frac{1}{18}$  and  $s(2, 3) = -\frac{1}{18}$ , and the leading term of  $\widehat{G}^3$  is 1. Hence the two cusps contribute  $\mathcal{P}_3(i)$  times

$$e\left(-\frac{i}{3}\right)e^{-\pi i/6} + e\left(-\frac{2i}{3}\right)e^{\pi i/6} = 2 \cos\left(\frac{2\pi i}{3} + \frac{\pi}{6}\right) = \sqrt{3}, -\sqrt{3}, 0 \quad (i \equiv 0, 1, 2 \pmod{3}),$$

that is,  $\sqrt{3}\chi(i)\mathcal{P}_3(i)$ , where  $e(x) = e^{2\pi i x}$ . (This is also confirmed by the exact agreement recorded in Remark 3.8.) For  $k \geq 6$  only the modulus 1 of each amplitude is used. Summing,  $E_{\text{con}} = E_{\text{arc}} + E_{\text{rem}} + E_{\text{ng}} = 50.1501$ .

*Certificate:* `sz_expansion_audit.py` (recomputes the constants); `log`  
`cubic/logs/03_sz_audit.log`.

□

#### 4. THE THIRD BORWEIN CONJECTURE

In this section  $p = 5$ ,  $\delta = 1$ ,  $P_n = P_{5,n} = (q; q)_{5n}/(q^5; q^5)_n$ ,  $c_n(m) = c(m) = [q^m]P_n$ ,  $\deg P_n = 10n^2$ ,  $N = 5n + 1$ ,  $a_5 = 2\pi^2/75$ ,  $\omega_5 = e^{2\pi i/5}$ ,  $s(i) = [q^i]G_5$ ,  $E_n = \prod_{k > 5n, 5 \nmid k} (1 - q^k)^{-1}$ ,  $\lambda = m/N$  and  $\mu = m/(10n^2)$ . The amplitude of the  $k = 5$  main term is

$$\mathbf{a}(i) = 2 \cos(2\pi i/5 + \pi/5) + 2 \cos(4\pi i/5) = \frac{5+\sqrt{5}}{2}, -\sqrt{5}, -\frac{5-\sqrt{5}}{2}, 0, 0 \quad (i \equiv 0, 1, 2, 3, 4).$$

It vanishes on the thin classes 3, 4 of Lemma 2.3(b). In the classes 0, 1, 2,  $s(m)$  has the non-vanishing main term  $\mathbf{a}(m)\mathcal{P}_5(m)$  (Theorem 4.3).

**Theorem 4.1.** *For every  $n \geq 1$  and every  $m$ ,  $c_n(m) \geq 0$  if  $5 \mid m$  and  $c_n(m) \leq 0$  otherwise. That is, the Third Borwein conjecture holds.*

Labels. Each step below is of one of the following kinds: a complete written argument; an inequality verified in ball arithmetic over its whole parameter range (“certified”; a *Certificate* line names the program and log); a verification by exact integer arithmetic (“exact”); or a standard fact (Appendix B).

**4.1. Statement, reduction and regions.** By Lemma 2.2 it suffices to prove the sign pattern for  $0 \leq m \leq 5n^2$ , that is, for  $0 \leq \mu \leq \frac{1}{2}$ . The three analytic regions are

- **band:**  $\lambda < 4$  (that is,  $m < 4N$ );
- **Laplace region:**  $\lambda \geq 4$  and  $\mu < 0.105$ ;
- **centre:**  $\mu \geq 0.105$ .

Their union is  $\{0 \leq m \leq 5n^2\}$  trivially. Table 3 summarises the covering; Theorem 4.35 checks that each region’s certificate covers the whole of its region for  $n \geq 980$ .

| region                        | classes | method                                                          | where  |
|-------------------------------|---------|-----------------------------------------------------------------|--------|
| $n \leq 979$                  | all     | exact integer computation                                       | §4.3   |
| $\lambda < 4$                 | 3, 4    | one-part block, exact to $n \leq 2000$ , majorant $\Psi$ beyond | §4.4.1 |
| $\lambda < 4$                 | 0, 1, 2 | exact to $n \leq 2000$ , per- $m$ inequality beyond             | §4.4.2 |
| $\lambda \geq 4, \mu < 0.105$ | 0, 1, 2 | saddle point, $V \in [2, 12]$ tiles and tail cell               | §4.5.3 |
| $\lambda \geq 4, \mu < 0.105$ | 3, 4    | re-centred (thin) saddle, tiles and tail cell                   | §4.5.4 |
| $\mu \geq 0.105$              | all     | near-peak Laplace, off-peak lemma, combination                  | §4.6   |

TABLE 3. The covering for  $p = 5$  (for  $n \geq 980$  in the analytic regions).

By Lemma 2.1,  $e(j)$  counts the partitions of  $j$  into parts  $> 5n$  that are not divisible by 5; for  $j < 4N$  at most three parts fit.

**Remark 4.2** (cusp by cusp). In the expansion of §4.2, the  $k = 10$  and  $k = 15$  amplitudes also vanish on  $i \equiv 3, 4 \pmod{5}$ , to 390 or more digits (`cusp10.py`, `cusp15b.py`). This is not used in the proof.

**4.2. The expansion of  $G_5$ .** *Notation for this subsection only.* The coefficient index is written  $\nu$ , and  $i = \sqrt{-1}$ .  $N$  is the Farey order, not  $5n + 1$ .  $e(x) = e^{2\pi i x}$ ;  $f(x) = F(x) = 1/(x; x)_\infty$ ;  $s(h, k) = \sum_{r=1}^{k-1} \frac{r}{k} \left( \left( \frac{hr}{k} \right) \right)$  is the Dedekind sum, and  $\omega(h, k) = e^{\pi i s(h, k)}$ ;  $t_0 = e^{-2\pi}$ .

4.2.1. *Statement.* Put  $B = 1/3$ ,  $C = C(\nu) = 2\nu - 1/3$ ,  $K(\nu) = \lfloor \sqrt{4\pi\nu} \rfloor + 2$ ,  $\mathcal{P}_k(\nu) = \frac{2\pi}{k} \sqrt{B/C} I_1\left(\frac{2\pi}{k} \sqrt{BC}\right)$  and  $\mathfrak{a}(\nu) = \sum_{h=1}^4 w_h \omega_5^{-\nu h}$ , where  $w_h = e^{-\pi i s(h,5)}$ .

**Theorem 4.3** (explicit expansion of  $G_5$ ). *For every  $\nu \geq 1$ ,*

$$|s(\nu) - \mathfrak{a}(\nu) \mathcal{P}_5(\nu)| \leq \sum_{5|k, 10 \leq k \leq K(\nu)} \varphi(k) \mathcal{P}_k(\nu) + E_{\text{arc}} + E_{\text{rem}} + E_{\text{ng}},$$

where

- $E_{\text{arc}} = \sqrt{2}\pi \cdot \frac{4}{25} e^{1+\pi/3} \leq 5.506447$ ;
- $E_{\text{rem}} = 2\sqrt{2} \cdot \frac{4}{25} e^{1+\pi/3} (f(t_0)f(t_0^5) - 1) \leq 0.006571$ , with  $t_0 = e^{-2\pi}$ ;
- $E_{\text{ng}} = 2\sqrt{2} e K_5 \leq 21.950524$ , with  $K_5 = \sqrt{5} e^{-\pi/15} f(e^{-2\pi/5}) f(e^{-2\pi}) \leq 2.8549955$ .

Their sum is  $E_{\text{con}} \leq 27.463542 \leq 27.4636$ .

Every  $\mathcal{P}_k(\nu)$  is positive:  $C \geq 5/3$ , and  $I_1 > 0$  on  $(0, \infty)$ . By §4.2.9,  $s(1,5) = 1/5$ ,  $s(2,5) = s(3,5) = 0$  and  $s(4,5) = -1/5$ . So  $w_1 = e^{-\pi i/5}$ ,  $w_2 = w_3 = 1$ ,  $w_4 = e^{\pi i/5}$ , and  $\mathfrak{a}(\nu) = 2 \cos(2\pi\nu/5 + \pi/5) + 2 \cos(4\pi\nu/5)$ , as above.

4.2.2. *Farey order.* Take  $N = K(\nu)$ . Two properties are used.

- **(N1)**  $N \geq 5$  for all  $\nu \geq 1$ , since  $\sqrt{4\pi} > 3$ . So the denominator  $k = 5$  always occurs. (The choice  $N = K(\nu) - 1$  fails at  $\nu = 1$ : it gives  $N = 4$ .)
- **(N2)**  $N + 1 = \lfloor \sqrt{4\pi\nu} \rfloor + 3 > \sqrt{4\pi\nu}$ . Hence  $2\pi C/(N+1)^2 = (4\pi\nu - 2\pi/3)/(N+1)^2 < 1$ .

4.2.3. *The Rademacher path.*

**Lemma 4.4** (Farey dissection; Fact B.2). *Take the Farey fractions of order  $N$  cyclically on  $[0, 1)$ , and let  $h_1/k_1 < h/k < h_2/k_2$  be neighbours. Let  $C(h/k)$  be the Ford circle with centre  $h/k + i/(2k^2)$  and radius  $1/(2k^2)$ , and let  $\gamma_{h,k}$  be its upper arc between its tangency points with  $C(h_1/k_1)$  and  $C(h_2/k_2)$ . Then these arcs join into a path from  $\tau_0$  to  $\tau_0 + 1$  in  $\mathbb{H}$ , and  $s(\nu) = \int_{\tau_0}^{\tau_0+1} G_5(e(\tau)) e(-\nu\tau) d\tau$ , taken over a horizontal segment, may be deformed onto this path by Cauchy's theorem. The integrand is holomorphic on  $\mathbb{H}$  and 1-periodic.*

The substitution  $\tau = h/k + iz/k^2$  maps  $C(h/k)$  onto  $K : |z - \frac{1}{2}| = \frac{1}{2}$ . Also  $d\tau = (i/k^2)dz$  and  $e(-\nu\tau) = e(-\nu h/k) e^{2\pi\nu z/k^2}$ . This gives

$$s(\nu) = \sum_{k \leq N} \sum_{\substack{0 \leq h < k \\ (h,k)=1}} \frac{i}{k^2} e(-\nu h/k) \int_{z'}^{z''} G_5(e(h/k + iz/k^2)) e^{2\pi\nu z/k^2} dz, \quad (5)$$

with  $z' = (k^2 + ikk_1)/(k^2 + k_1^2)$  and  $z'' = (k^2 - ikk_2)/(k^2 + k_2^2)$ . The integral runs clockwise along  $K$  from  $z'$  ( $\text{Im} > 0$ ) through  $z = 1$  to  $z''$  ( $\text{Im} < 0$ ). The tangency point of  $C(h/k)$  with  $C(h_1/k_1)$  is  $h/k - k_1/(k(k^2 + k_1^2)) + i/(k^2 + k_1^2)$ , and it maps to  $z'$ ; the same holds for  $z''$ .

*The arc of  $0/1$ .* In the cyclic dissection of  $[0, 1)$ , the circle  $C(0/1)$  contributes two pieces: the arc from  $\tau_0 = i$  to its tangency with  $C(1/N)$ , and the arc of  $C(1/1) = C(0/1) + 1$  from its tangency with  $C((N-1)/N)$  to  $\tau_0 + 1 = i + 1$ . The integrand  $G_5(e(\tau)) e(-\nu\tau)$  is 1-periodic. Translate the second piece by  $-1$ : it becomes the arc of  $C(0/1)$  from the tangency with  $C(-1/N)$  to  $i$ . The two pieces then join into a single arc of  $C(0/1)$ , from the tangency with  $C(-1/N)$  through  $i$  to the tangency with  $C(1/N)$ . In the  $z$ -variable ( $h = 0, k = 1, \tau = iz$ ), this is the arc of  $K$  from  $z'$  (with  $k_1 = N$ ) through  $z = 1$  (that is,  $\tau = i$ ) to  $z''$  (with  $k_2 = N$ ). So the term  $h/k = 0/1$  in (5) is one Rademacher arc like every other, and Lemma 4.5 applies to it with  $k_1 = k_2 = N$ . (These neighbours satisfy  $k + k_j = N + 1$ .)

**Lemma 4.5** (geometry). (b)  $|z'|, |z''| \leq \sqrt{2}k/(N+1)$ , and  $\text{Re } z', \text{Re } z'' \leq 2k^2/(N+1)^2$ .

- (c) On the chord  $s_{h,k} = [z', z'']$ :  $\operatorname{Re}(1/z) \geq 1$ , and  $0 < \operatorname{Re} z \leq 2k^2/(N+1)^2$ . Also  $|s_{h,k}| \leq 2\sqrt{2}k/(N+1)$ .
- (d) On the arcs of  $K$  from  $z''$  to 0 and from 0 to  $z'$  (the complement of the Rademacher arc):  $\operatorname{Re}(1/z) = 1$  and  $\operatorname{Re} z \leq 2k^2/(N+1)^2$ . Each of these arcs has length at most  $(\pi/2)\sqrt{2}k/(N+1)$ .

*Proof.* (b) Farey neighbours satisfy  $k + k_j \geq N+1$  (Fact B.3). So  $k^2 + k_j^2 \geq (N+1)^2/2$ . Therefore  $|z'| = k/\sqrt{k^2 + k_1^2} \leq \sqrt{2}k/(N+1)$  and  $\operatorname{Re} z' = k^2/(k^2 + k_1^2) \leq 2k^2/(N+1)^2$ . The same holds for  $z''$ .

(c) The map  $z \mapsto 1/z$  sends  $K \setminus \{0\}$  onto the line  $\operatorname{Re} w = 1$ , and the open disc onto  $\operatorname{Re} w > 1$ . The chord lies in the closed disc and avoids 0.  $\operatorname{Re} z$  is affine on the chord and positive at both ends. Finally,  $|s_{h,k}| \leq |z'| + |z''|$ .

(d)  $\operatorname{Re}(1/z) = 1$  on  $K \setminus \{0\}$ . Let  $\theta \in (0, \pi]$  be the central angle of the arc from  $z'$  to 0. Its chord has length  $|z'| = \sin(\theta/2)$  and its arc length is  $\theta/2 \leq (\pi/2)\sin(\theta/2)$ , by Jordan's inequality. On a semicircle,  $\operatorname{Re} z = (1 + \cos \psi)/2$  is monotone, so  $\operatorname{Re} z \leq \operatorname{Re} z'$  on this arc. The arc from  $z''$  is handled the same way.  $\square$

**Corollary 4.6.** *On every chord and every arc in Lemma 4.5(d),  $|e^{\pi Cz/k^2}| \leq e^{2\pi C/(N+1)^2} < e$ .*

*Proof.* Lemma 4.5 and (N2).  $\square$

**Remark 4.7** (the regions of deformation). Two facts about the region  $\Delta_{h,k}$  bounded by the Rademacher arc  $z' \rightarrow 1 \rightarrow z''$  and the chord  $s_{h,k}$  are used below.

- $\Delta_{h,k}$  lies in the half-plane  $\operatorname{Re} z \geq \min(\operatorname{Re} z', \operatorname{Re} z'') > 0$ . On each of the two semicircular pieces of the arc,  $\operatorname{Re} z$  is monotone, with its maximum at  $z = 1$ . So on the arc,  $\operatorname{Re} z \geq \min(\operatorname{Re} z', \operatorname{Re} z'')$ . On the chord this holds by affinity. The region  $\Delta_{h,k}$  is a circular segment, which is the convex hull of the arc, so it lies in that half-plane as well.
- The integrands are holomorphic on  $\Delta_{h,k}$ . In particular  $\Delta_{h,k}$  avoids  $z = 0$ . For  $\operatorname{Re} z > 0$ ,  $\tau = h/k + iz/k^2$  has  $\operatorname{Im} \tau = \operatorname{Re} z/k^2 > 0$ , so  $G_5(e(\tau))$  and the functions derived from it in §4.2.4 are holomorphic there.

Cauchy's theorem therefore allows every integral over a Rademacher arc to be replaced by the integral over its chord. This is used in §4.2.6 and §4.2.7 alike.

#### 4.2.4. The modular transformation at a cusp.

**Fact 4.8** (Fact B.1). *Let  $k \geq 1$ ,  $(h, k) = 1$ ,  $\operatorname{Re} z > 0$ , and  $hH \equiv -1 \pmod{k}$ . Put  $x = \exp(2\pi i h/k - 2\pi z/k^2)$  and  $x' = \exp(2\pi i H/k - 2\pi/z)$ . Then*

$$F(x) = \omega(h, k)(z/k)^{1/2} \exp\left(\frac{\pi}{12z} - \frac{\pi z}{12k^2}\right) F(x'),$$

*with the principal square root. Equivalently,  $(x; x)_\infty = \omega(h, k)^{-1}(z/k)^{-1/2} \exp(-\frac{\pi}{12z} + \frac{\pi z}{12k^2})(x'; x')_\infty$ .*

Let  $x = e(\tau)$  with  $\tau = h/k + iz/k^2$ . Then  $G_5(x) = (x; x)_\infty F(x^5)$ .

Case  $5 \mid k$  (growing cusps),  $k = 5k_1$ . Write  $x^5 = \exp(2\pi i h/k_1 - 2\pi z_1/k_1^2)$  with  $z_1 = z/5$ , and apply Fact 4.8 to  $(h, k_1, z_1)$ . Then

$$G_5(x) = \Phi_{h,k} \exp\left(\frac{\pi}{3z} - \frac{\pi z}{3k^2}\right) \hat{G}_{h,k}(z), \quad (6)$$

where  $\Phi_{h,k} = \omega(h, k_1)/\omega(h, k)$  is unimodular and  $\hat{G}_{h,k} = (x'; x')_\infty F(x'_5)$ , with  $x'_5 = \exp(2\pi i H_1/k_1 - 10\pi/z)$ . The square roots cancel exactly, because  $(z_1/k_1)^{1/2} = (z/k)^{1/2}$  is the same principal root; so no branch choice enters. For the exponents:  $-\frac{\pi}{12z} + \frac{5\pi}{12z} = \frac{\pi}{3z}$  and  $\frac{\pi z}{12k^2} - \frac{5\pi z}{12k^2} = -\frac{\pi z}{3k^2}$ . For  $k = 5$ :  $\Phi_{h,5} = \omega_h$ .

Case  $5 \nmid k$  (non-growing cusps). Apply Fact 4.8 to  $(5h, k, 5z)$  with  $5hH_5 \equiv -1 \pmod{k}$ . Then

$$G_5(x) = \Phi'_{h,k} \sqrt{5} \exp\left(-\frac{\pi}{15z} - \frac{\pi z}{3k^2}\right)(x'; x')_\infty F(x''_5), \quad x''_5 = \exp(2\pi i H_5/k - 2\pi/(5z)), \quad (7)$$

with  $|\Phi'_{h,k}| = 1$ . Here  $(z/k)^{-1/2}(5z/k)^{1/2} = \sqrt{5}$ , because  $z$  and  $5z$  have the same argument.

The shift in  $C$ . In (5), the factor  $e^{-\pi z/(3k^2)}$  combines with  $e^{2\pi\nu z/k^2}$  to give  $e^{\pi C z/k^2}$ , where  $C = 2\nu - 1/3$ .

Formulae (6) and (7) were checked against direct  $q$ -series, phases included: 8 cusps at 60 digits (relative error  $\leq 4 \cdot 10^{-60}$ ), and independently by the referee of this section at 10 cusps and two values of  $z$  (relative error  $\leq 4.6 \cdot 10^{-49}$ ).

Majorant of  $\hat{G}$ . Put  $u = e^{-2\pi/z}$  and  $w = \operatorname{Re}(1/z)$ . Expanding  $(x'; x')_\infty = \prod(1 - e(mH/k)u^m)$  and  $F(x'_5) = \prod(1 - e(mH_1/k_1)u^{5m})^{-1}$  gives  $\hat{G} = 1 + \sum_{j \geq 1} g_j u^j$ . Coefficientwise,  $|g_j| \leq [y^j] \prod(1 + y^m)f(y^5) \leq [y^j]f(y)f(y^5) =: \phi_j$ , because distinct partitions are partitions. Hence

$$|\hat{G} - 1| \leq \sum_{j \geq 1} \phi_j e^{-2\pi j w}. \quad (8)$$

4.2.5. Growing cusps: main term. By (5) and (6), the cusp  $h/k$  with  $5 \mid k$  contributes  $\frac{i}{k^2} e(-\nu h/k) \Phi_{h,k}[M_{h,k} + R_{h,k}]$ , where

$$M_{h,k} = \int_{z'}^{z''} e^{\pi/(3z) + \pi C z/k^2} dz, \quad R_{h,k} = \int_{z'}^{z''} e^{\pi/(3z) + \pi C z/k^2} (\hat{G}_{h,k} - 1) dz.$$

**Fact 4.9** (Bessel integral; Fact B.4). For  $x > 0$  and  $c > 0$ ,  $I_1(x) = \frac{x/2}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{s+x^2/(4s)} s^{-2} ds$ .

From the Hankel loop to the vertical line. DLMF 10.9.19 [5] gives  $I_1(x) = \frac{x/2}{2\pi i} \int_{-\infty}^{(0+)} e^{s+x^2/(4s)} s^{-2} ds$ , over a Hankel loop that starts and ends at  $-\infty$  and encircles the negative real axis. For integer order the only singularity of the integrand is the essential singularity at  $s = 0$ ; the loop has no branch cut to respect. Fix  $R > c$  and close the region between the loop and the line  $\operatorname{Re} s = c$  with the two arcs  $\Gamma_R^\pm$  of  $|s| = R$  in  $\{\operatorname{Re} s \leq c\}$ , running from  $c \pm i\sqrt{R^2 - c^2}$  to the loop. The region enclosed contains no singularity, so by Cauchy's theorem the loop integral equals the line integral truncated at height  $\sqrt{R^2 - c^2}$ , plus the integrals over  $\Gamma_R^\pm$  (and minus the loop's own parts beyond  $|s| = R$ ). On  $\Gamma_R^\pm$ :  $|e^s| \leq e^c$ ;  $|e^{x^2/(4s)}| \leq e^{x^2/(4R)} \leq e^{x^2/(4c)}$ ;  $|s^{-2}| = R^{-2}$ ; the total length is at most  $2\pi R$ . So the arcs contribute  $O(R^{-1}) \rightarrow 0$ . The loop's parts beyond  $|s| = R$  vanish too, since  $e^s$  decays along the negative axis. The line integral converges absolutely, because the integrand is  $O(|s|^{-2})$  on it. Letting  $R \rightarrow \infty$  proves Fact 4.9.

**Lemma 4.10.** Let  $K$  be oriented clockwise,  $B > 0$ ,  $C > 0$ . Then  $\frac{i}{k^2} \oint_K e^{\pi B/z + \pi C z/k^2} dz = \mathcal{P}_k$ .

*Proof.* Put  $w = 1/z = 1 + i\rho$ ,  $\rho \in \mathbb{R}$ . The clockwise traversal of  $K \setminus \{0\}$  corresponds to  $\rho$  running from  $-\infty$  to  $\infty$ . With  $dz = -dw/w^2$ , the integral becomes  $-\int_{1-i\infty}^{1+i\infty} e^{\pi B w + \pi C/(k^2 w)} w^{-2} dw$ . Substituting  $s = \pi B w$  gives  $-\pi B \int e^{s+x^2/(4s)} s^{-2} ds$ , with  $x = 2\pi\sqrt{BC}/k$ . By Fact 4.9 this equals  $-\pi B \cdot 2\pi i \cdot (2/x) I_1(x)$ . Multiplying by  $i/k^2$  gives  $4\pi^2 B I_1(x)/(k^2 x) = \mathcal{P}_k$ .  $\square$

Numerical check at  $k = 10$ ,  $\nu = 7$ : 0.08174002496770177 on both sides; the referee's check agrees to 30 digits. (Lemma 4.10 is also the identity used for  $p = 3$  in §3.6, with  $B = \frac{1}{2}$ .)

Completing the arc. On  $K \setminus \{0\}$  the integrand of  $M_{h,k}$  has modulus  $e^{\pi/3} |e^{\pi C z/k^2}|$ , which is bounded, so  $\oint_K$  converges absolutely. The complement of the Rademacher arc is the arc  $z'' \rightarrow 0 \rightarrow z'$ . By Lemma 4.5(d) and Corollary 4.6, the integrand there is at most  $e^{1+\pi/3}$ , and the total length is at most  $\pi\sqrt{2}k/(N+1)$ . Hence

$$\frac{i}{k^2} M_{h,k} = \mathcal{P}_k(\nu) + \rho_{h,k}, \quad |\rho_{h,k}| \leq \frac{\pi\sqrt{2}e^{1+\pi/3}}{k(N+1)}. \quad (9)$$

*Amplitudes.* Put  $\mathbf{a}_k(\nu) = \sum_{(h,k)=1} \Phi_{h,k} e(-\nu h/k)$  for  $5 \mid k$ . This is a sum of  $\varphi(k)$  unimodular terms, so  $|\mathbf{a}_k| \leq \varphi(k)$ , and  $\mathbf{a}_5 = \mathbf{a}$ .

4.2.6. *Growing cusps: remainder.* By Remark 4.7,  $R_{h,k}$  may be taken along the chord, where  $w \geq 1$ . By (8),  $|e^{\pi/(3z)}(\hat{G} - 1)| \leq \sum_{j \geq 1} \phi_j e^{(\pi/3 - 2\pi j)w}$ . Each exponent is negative, so each term decreases in  $w$ ; this is where  $B = 1/3 < 2$  is used. The sum is therefore at most its value at  $w = 1$ , namely  $e^{\pi/3}(f(t_0)f(t_0^5) - 1)$ . With the chord length and Corollary 4.6,

$$\left| \frac{i}{k^2} R_{h,k} \right| \leq \frac{1}{k^2} \cdot \frac{2\sqrt{2}k}{N+1} e^{1+\pi/3} (f(t_0)f(t_0^5) - 1). \quad (10)$$

The  $j = 1$  term decays, so no correction term needs to be extracted.

4.2.7. *Non-growing cusps.* Here  $5 \nmid k$ . By Remark 4.7 the integral may be moved to the chord. By (7), for  $w \geq 1$  on the chord:  $|(x'; x')_\infty| \leq f(e^{-2\pi w}) \leq f(e^{-2\pi})$ ;  $|F(x''_5)| \leq f(e^{-2\pi w/5}) \leq f(e^{-2\pi/5})$ ;  $e^{-\pi w/15} \leq e^{-\pi/15}$ . Each bound is non-increasing in  $w$ . With Corollary 4.6, the integrand is at most  $K_5 e$ , so each such cusp contributes at most

$$\frac{1}{k^2} \cdot \frac{2\sqrt{2}k}{N+1} \cdot e K_5. \quad (11)$$

4.2.8. *Assembly.* By (5) and (9)–(11),  $s(\nu) - \mathbf{a}(\nu)\mathcal{P}_5(\nu) - \sum_{5 \mid k, 10 \leq k \leq N} \mathbf{a}_k \mathcal{P}_k = E(\nu)$ , where

$$|E(\nu)| \leq \sum_{5 \mid k \leq N} \frac{\varphi(k) [\pi\sqrt{2} + 2\sqrt{2}(f(t_0)f(t_0^5) - 1)] e^{1+\pi/3}}{k(N+1)} + \sum_{5 \nmid k \leq N} \frac{\varphi(k) 2\sqrt{2}e K_5}{k(N+1)}.$$

For  $5 \mid k$ :  $\varphi(k)/k \leq 4/5$ , and there are at most  $N/5$  such  $k$ , so the first weight sum is less than  $4/25$ . For  $5 \nmid k$ :  $\varphi(k)/k \leq 1$ , and there are at most  $N$  values of  $k$ , so the second weight sum is less than 1. This gives  $|E| \leq E_{\text{arc}} + E_{\text{rem}} + E_{\text{ng}}$ . Using  $|\mathbf{a}_k| \leq \varphi(k)$  and  $N = K(\nu)$  proves Theorem 4.3.  $\square$

4.2.9. *Constants.* Computed at 60 digits and independently at 50 digits by the referee:

| constant         | value               | rounded up |
|------------------|---------------------|------------|
| $K_5$            | 2.85499540499087... | 2.8549955  |
| $E_{\text{arc}}$ | 5.50644686778894... | 5.506447   |
| $E_{\text{rem}}$ | 0.00657086323166... | 0.006571   |
| $E_{\text{ng}}$  | 21.9505238422352... | 21.950524  |
| $E_{\text{con}}$ | 27.4635415732558... | 27.4636    |

The Dedekind sums  $s(1, 5) = \frac{1}{5}$ ,  $s(2, 5) = s(3, 5) = 0$  and  $s(4, 5) = -\frac{1}{5}$  were checked at 60 digits.

4.2.10. *Numerical confirmation and tightness (not used in the proof).* The referee checked Theorem 4.3 exhaustively over  $\nu \in [1, 5000]$  and at 1509 further  $\nu \leq 200000$ : no violation, maximum LHS/RHS 0.9045085. An earlier ball-arithmetic survey covered every  $\nu \leq 60000$  and every 7th  $\nu \leq 200000$  with no failures. The ratio tends to  $(5 + \sqrt{5})/8 = 0.9045085$  in the class  $\nu \equiv 2 \pmod{5}$ , where  $|\mathbf{a}_{10}| = (5 + \sqrt{5})/2$  is set against  $\varphi(10) = 4$ ; so the  $\sum \varphi(k) \mathcal{P}_k$  term must not be weakened in any later use.

*Uses of Theorem 4.3.* Write  $R(i)$  for its right-hand side. The certificates use  $|s(i) - \mathbf{a}(i)\mathcal{P}_5(i)| \leq R(i)$  for every  $i \geq 1$ , and exact values of  $s(i)$  for  $i \leq 200000$  (`g5_coef.py`  $\rightarrow$  `g5_coeffs.pkl`, cross-checked by an independent recomputation in `bandfix_coeffs.py`).

### 4.3. Exact computation.

**Theorem 4.11** (exact). *The conjecture holds, for every coefficient and in every class, for all  $n \leq 979$ .*

*Method.* The script works modulo  $x^{5N_B^2+1}$  and builds  $P_{N_A}$  from  $P_0 = 1$ . It obtains each next  $P_n$  by four exact shift-subtracts  $P \leftarrow P - x^k P$ , for  $k = 5n+1, \dots, 5n+4$  with  $5 \nmid k$ . Truncation is exact, because each factor feeds only higher degrees. It then checks every coefficient with  $0 \leq m \leq 5n^2$ , in every class; a strict wrong sign counts as a violation, zeros are allowed. The range is run in the segments 1–150, 150–291, 291–410, 410–580, 580–820, 820–920 and 920–1000, all with 0 violations; the last run was killed for lack of memory after the lines  $n = 920, \dots, 979$ , all with `wrong_sign=0`. The segments abut or overlap, so the exact range is  $1 \leq n \leq 979$ . Zeros occur only at  $m \lesssim 5n$ , i.e.  $\mu < 2 \cdot 10^{-3}$ , inside the band, where the conjectured signs are non-strict. Palindromy (Lemma 2.2) extends each check from  $m \leq 5n^2$  to all  $m$ .

*Certificate:* `centre3_exact.py N_A N_B; log third/logs/exact_1_150.log,`  
`exact_150_291.log, centre3_exact_*.log.`

□

*Further exact checks (used in §4.4).*

- Every class and every  $m < 4N$ , for every  $n \in [291, 2000]$ . The method is  $c_n = G_5 E_n \bmod q^{4N}$  with  $6E_n = 6 + 6p_1 + 3(p_1^2 + p_1(q^2)) + (p_1^3 + 3p_1(q^2)p_1 + 2p_1(q^3))$  (the cycle index of  $S_2$  and  $S_3$ ), where  $p_1 = \sum_{N \leq k < 4N, 5 \nmid k} q^k$ , computed with `fmpz_poly.mul_low`. The script was validated against the full product for  $n \leq 14$ , and it reports zero sign failures. An audit recomputed 16 values of  $n$  in  $[291, 2000]$  by a direct product that does not use  $E_n$  or the coefficient table, with identical results for all  $m < 4N$ .

*Certificate:* `bandfix_exact.py 291 2000; log third/logs/bandfix_exact.log.`

- The coefficients  $s(i)$  for  $i \leq 200000$  come from `g5_coef.py` (pentagonal series times  $p(k)$  at  $q^5$ ). They were recomputed independently.

*Certificate:* `bandfix_coeffs.py; log third/logs/bandfix_coeffs.log.`

**Remark 4.12.** The analytic certificates below hold from  $n \geq 821$  (Laplace classes 3–4 and the centre) or from  $n \geq 291$  (Laplace classes 0–2). The overlap with the exact range is therefore  $821 \leq n \leq 979$ . From here on,  $n \geq 980$ .

**4.4. The band,  $\lambda < 4$ .** Every part of  $E_n$  is  $\geq N$ . So for  $m < 4N$  at most three parts contribute, and  $c(m) = s(m) + b_1 + b_2 + b_3$ , where  $b_j$  collects the  $j$ -part terms. For  $j \in [N, 2N)$ ,  $e(j) = 1$  if  $5 \nmid j$  and  $e(j) = 0$  otherwise.

**4.4.1. Classes 3 and 4. One-part block.** Let  $m \equiv 3$  or  $4 \pmod{5}$  and write  $m = 5M + r$ . Then  $s(m) = 0$ , and  $b_1 = \sum_{j \in [N, m], 5 \nmid j} s(m - j) = \sum_{i \leq m - N, i \not\equiv m} s(i)$ . Since  $s(i) = 0$  for  $i \equiv 3, 4$ , only  $i \equiv 0, 1, 2$  survive. Also  $m - N = 5(M - n) + r - 1$  with  $r - 1 \in \{2, 3\}$ , so every block  $\{5t, 5t + 1, 5t + 2\}$  with  $t \leq T := M - n$  is included in full. Hence

$$b_1 = G(T) = \sum_{t \leq T} g(t), \quad g(t) = s(5t) + s(5t + 1) + s(5t + 2).$$

**Lemma 4.13.** *The main term satisfies  $\text{main}(g(t)) \leq -5\mathcal{P}'_5(5t)$ . Moreover  $g(t) < -4\mathcal{P}'_5(5t) < 0$  for  $t \geq 200$ .*

*Proof.* The main term of  $g(t)$  is  $\mathfrak{a}(0)\mathcal{P}_5(5t) + \mathfrak{a}(1)\mathcal{P}_5(5t + 1) + \mathfrak{a}(2)\mathcal{P}_5(5t + 2)$ . Since  $\mathfrak{a}(0) = |\mathfrak{a}(1)| + |\mathfrak{a}(2)|$ , this equals  $-|\mathfrak{a}(1)|[\mathcal{P}_5(5t + 1) - \mathcal{P}_5(5t)] - |\mathfrak{a}(2)|[\mathcal{P}_5(5t + 2) - \mathcal{P}_5(5t)]$ . By the series of  $I_1$ ,  $\mathcal{P}_5(i) = \sum_{r \geq 1} a_5^r x^{r-1} / (r! (r-1)!)$  is a power series with positive coefficients in  $x = i - \frac{1}{6}$  (not

in  $i$  itself). So  $\mathcal{P}'_5$  is increasing for  $x > 0$ , and the main term is at most  $-(|a(1)| + 2|a(2)|)\mathcal{P}'_5(5t) = -5\mathcal{P}'_5(5t)$ .

By Theorem 4.3,  $|g(t) - \text{main}| \leq 3R_{\text{grp}}(5t + 2)$ , where  $R_{\text{grp}}$  is the (increasing) right-hand side of Theorem 4.3. Lemma L1 states  $3R_{\text{grp}}(X + 2) \leq 0.018\mathcal{P}'_5(X)$  for all  $X \geq 1000$ . Its proof has four steps:

- (1) write  $\mathcal{P}'_5(X) = 4a_5^2 I_2(Y)/Y^2$  with  $Y = 2\sqrt{a_5(X - 1/6)}$ ;
- (2) use  $I_2 \geq I_{5/2}$  (monotonicity in the order, Fact B.5) and the closed form of  $I_{5/2}$ , which give  $I_2(Y) \geq e^Y(1 - 3/Y - 2e^{-2Y})/\sqrt{2\pi Y}$ ;
- (3) use  $I_1 \leq I_{1/2}$ ,  $\mathcal{P}_k \leq \mathcal{P}_{10}$  for  $k \geq 10$ , and  $\sum_{5|k \leq K} \varphi(k) \leq K^2/10 + K/2$ ;
- (4) bound the ratio by  $c_1 X^{3/2} e^{-Y/2} + c_2 Y^{5/2} e^{-Y}$ . Both terms decrease for  $X \geq 1000$  (sign of the log-derivative), so the bound evaluated at  $X = 1000$  in Arb holds for all larger  $X$ : the log prints L1:  $\sup_{\{X \geq 1000\}} 3 R_{\text{grp}}(X+2)/\mathcal{P}'_5(X) \leq [0.017972 \text{ +/- } 1.12\text{e-}7]$ .

This gives  $g(t) \leq -5\mathcal{P}'_5 + 0.018\mathcal{P}'_5 < -4\mathcal{P}'_5$  for  $5t \geq 1000$ .

Certificate: `bandfix_analytic.py 10001; log third/logs/bandfix_analytic.log`. □

**Lemma 4.14** (exact). *For  $t \leq 39999$ ,  $g(t) < 0$  except at  $g(1) = 1$  and  $g(5) = 0$ . Every partial sum satisfies  $G(T) \leq 0$ , with equality only at  $T = 1$ . The computation uses the table of exact  $s(i)$ , which the same script recomputes independently first.*

Certificate: `bandfix_coeffs.py; log third/logs/bandfix_coeffs.log`.

**Corollary 4.15.**  *$G(T) \leq 0$  for every  $T$ , and  $G(T) \leq g(T) < 0$  for  $T \geq 200$ . Hence, for classes 3 and 4 and every  $n$ : for  $m < N$ ,  $c(m) = s(m) = 0$ ; for  $N \leq m < 2N$ ,  $c(m) = G(T) \leq 0$ .*

*The range  $2N \leq m < 4N$ .*

- For  $980 \leq n \leq 2000$  (exact): by the exact band run, which covers  $291 \leq n \leq 2000$ . Over that whole run the log prints the largest ratio  $|b_2 + b_3|/|b_1|$  as  $0.0001$ , at  $n = 291$ ,  $m = 5819$ ; an audit's rerun gives the unrounded value  $1.1302 \cdot 10^{-4}$ .
- For  $n > 2000$  (so  $N \geq 10001$ ). Write  $I = m - 2N$ . Then  $5T \geq I + N - 3$ .
  - Denominator.  $|b_1| \geq |g(T)| \geq 4\mathcal{P}'_5(I + N - 3)$ , by Corollary 4.15 and the monotonicity of  $\mathcal{P}'_5$ .
  - Numerator.  $|b_2| + |b_3| \leq \sum_{d \leq I} w(2N + d)|s(I - d)|$ . Here the number of pairs is at most  $d/2 + 1$ , and the number of triples at  $3N + e$  is at most  $(e + 3)^2/12 + 1/2$ . Also  $|s(i)| \leq e^{2\sqrt{a_5 i}}$ : this is exact for  $i \leq 200000$ , and beyond that Theorem 4.3 gives the stronger constant  $7.7 \cdot 10^{-5}$ .
  - Split. Split the sum at  $D = \lfloor 3\sqrt{2N/a_5} \log N \rfloor$ . The exponent  $2\sqrt{a_5}(\sqrt{I} - \sqrt{I + N - 19/6})$  increases in  $I$ , so it is largest at  $I = 2N$ . This gives

$$\frac{|b_2| + |b_3|}{|b_1|} \leq \Psi(N) = C_\Psi N^{9/4} (\log N)^2 e^{-2\sqrt{a_5}(\sqrt{3}-\sqrt{2})\sqrt{N}}.$$

- Conclusion.  $d \log \Psi / dN < 0$  exactly when  $(c_c/2)\sqrt{N} > 9/4 + 2/\log N$ , where  $c_c = 2\sqrt{a_5}(\sqrt{3} - \sqrt{2})$  and  $c_c/2 = 0.16304$ . This first holds at  $N = 257$  and persists for larger  $N$ , so  $\Psi$  is decreasing for  $N \geq 257$ . In Arb,  $\Psi(10001) \leq 0.19769$ . Hence  $c(m) \leq b_1(1 - 0.1977) < 0$ .

Certificate: `bandfix_analytic.py 10001; log third/logs/bandfix_analytic.log`.

**Corollary 4.16.** *For classes 3 and 4, every  $n \geq 980$  and every  $m < 4N$ :  $c(m) \leq 0$ .*

4.4.2. *Classes 0, 1, 2.* For  $980 \leq n \leq 2000$  the band is covered by the exact run. For  $n > 2000$  write  $c(m) = s(m) + \sum_{j \geq N} e(j)s(m-j)$ . The weights satisfy  $e(j) \leq 1$  for  $j < 2N$  and  $e(j) \leq 1 + (N+1) + ((N+3)^2/12 + 1/2) \leq N^2 \leq m^2$  for  $j < 4N$ . With  $A(I) = \sum_{i \leq I} |s(i)|$ , this gives

$$|c(m) - s(m)| \leq A(m-N) + m^2 A(m-2N) \leq A(m-N_{\min}) + m^2 A(m-2N_{\min}),$$

where  $N_{\min}(m) = \max(10001, \lfloor m/4 \rfloor + 1)$ . The second inequality uses that  $A$  is non-decreasing and that  $m < 4N$  is equivalent to  $N \geq \lfloor m/4 \rfloor + 1$ . So one inequality per  $m$  covers every  $n > 2000$  at once.

**Lemma 4.17.** (A) (exact,  $m \leq 200000$ .)  $\operatorname{sgn} s(m) = \operatorname{sgn} a(m)$  and  $|s(m)| > A(m-N_{\min}) + m^2 A(m-2N_{\min})$  for every  $m$  in classes 0, 1, 2, except  $s(7) = 0$ . That coefficient has  $m < N$ , so  $c(7) = 0$ , which is allowed. The worst ratio is  $5.06 \cdot 10^{-10}$ .

(B) (certified,  $m > 200000$ .) With the inputs  $|s(i)| \leq e^{c\sqrt{i}}$  ( $c = 2\sqrt{a_5}$ ),  $\sum_{i \leq I} e^{c\sqrt{i}} \leq e^{c\sqrt{I}}(1 + 2\sqrt{I}/c)$ ,  $\mathcal{P}_5(m) \geq h_0 m^{-3/4} e^{c\sqrt{m}}$  (via  $I_1 \geq I_{3/2}$ ) and  $R(m) \leq 1.3 m e^{c\sqrt{m}/2} + E_{\text{con}}$ , one has  $[\text{corrections} + R(m)]/((5 - \sqrt{5})/2 \cdot \mathcal{P}_5(m)) \leq Q \leq 5.14 \cdot 10^{-20}$ . Each term has the form  $C' m^{p'} e^{-\kappa\sqrt{m}}$  with  $p' \in \{1.25, 3.25, 1.75, 0.75\}$ , and is decreasing once  $\sqrt{m} > 2p'/\kappa$ .

Certificate: `cls012_band.py`; log `third/logs/cls012_band.log`.

**Corollary 4.18.** For classes 0, 1, 2, every  $n \geq 980$  and every  $m < 4N$ ,  $c(m)$  has the conjectured sign.

4.5. **The Laplace region**,  $\lambda \geq 4$ ,  $\mu < 0.105$ . Both certificates of this subsection share one exact representation (§4.5.1) and one Euler–Maclaurin lemma (§4.5.2). They differ in the saddle, because classes 3 and 4 are thin. Write  $\varpi_\eta = 1 + i\eta$ .

4.5.1. *Exact Bessel–Laplace representation.* Write  $a(i) = \sum_{h=1}^4 A_h \omega_5^{hi}$  with  $A_1 = e^{i\pi/5}$ ,  $A_4 = e^{-i\pi/5}$  and  $A_2 = A_3 = 1$ . By Lemma 2.1 and Theorem 4.3,

$$c(m) = e(m) + \sum_h A_h \omega_5^{hm} J_h + \rho, \quad J_h = \sum_{j < m} e(j) \omega_5^{-hj} \mathcal{P}_5(m-j), \quad |\rho| \leq \sum_{j < m} e(j) R(m-j).$$

Here  $e(m) = e(m)s(0)$  is the  $j = m$  term, and  $\omega_5^{-hj} \omega_5^{hm} = \omega_5^{h(m-j)}$  reproduces  $a(m-j)$ .

For  $x = i - 1/6 > 0$  we have  $\mathcal{P}_5(i) = \sqrt{a_5/x} I_1(2\sqrt{a_5 x})$ , which is the inverse Laplace transform of  $e^{a_5/u} - 1$ . Put  $r_1(u) = e^{a_5/u} - 1 - a_5/u = O(|u|^{-2})$ . Then  $\frac{1}{2\pi i} \int_{(W)} r_1(u) e^{xu} du = \mathcal{P}_5 - a_5$  for  $x > 0$  and  $= 0$  for  $x < 0$ . Since  $r_1 = O(|u|^{-2})$ , Fubini applies, and for every  $W > 0$ , exactly,

$$J_h = a_5 H_h + K_h, \quad H_h = \sum_{j \leq m-1} e(j) \omega_5^{-hj},$$

$$K_h = \frac{1}{2\pi} \int_{\mathbb{R}} r_1(W + iy) e^{(m-1/6)(W+iy)} E_n(\omega_5^{-h} e^{-W-iy}) dy.$$

4.5.2. *Euler–Maclaurin lemma for the twisted  $E_n$ .*

**Lemma 4.19.** Let  $\theta \in (0, 1)$ , and let  $f$  be such that  $f, f' \rightarrow 0$  at  $\infty$  and  $f''$  is integrable. Then

$$\sum_{t \geq n} f(t + \theta) = \int_n^\infty f - B_1(\theta) f(n) + R, \quad |R| \leq \frac{1}{12} (|f'(n)| + \int_n^\infty |f''|).$$

*Proof.* On each unit interval  $[t, t+1]$ , write  $f(t + \theta) - \int_t^{t+1} f = \int_t^{t+1} k(x-t) f'(x) dx$ , where  $k(x) = x - H(x - \theta)$  and  $H$  is the Heaviside function. The mean of  $k$  is  $B_1(\theta)$ , and  $k - B_1(\theta) = \tilde{B}_1(x - \theta)$ . The mean part telescopes to  $-B_1(\theta) f(n)$ . Integrate the remaining part by parts once more, with kernel  $P(x) = (\tilde{B}_2(x - \theta) - B_2(\theta))/2$ . Its constant part contributes  $(B_2(\theta)/2) f'(n)$ ,

with  $|B_2(c/5)|/2 \leq 11/300 < 1/12$ . Its periodic part is bounded by  $\sup |\tilde{B}_2|/2 = 1/12$  times  $\int |f''|$ .  $\square$

The constant part must be split off: without the split,  $\sup |P|$  reaches  $0.12 > 1/12$  at  $\theta = 2/5$ .

*Application.* Write  $k = 5t + c$  with  $c \in \{1, 2, 3, 4\}$ . Put  $\Phi_c(s) = \log(1 - \omega_5^{-hc}e^{-s})$ ,  $u = W\varpi_\eta$ ,  $\sigma = 5nu$ ,  $V = 5nW$  and  $\xi = e^{-V}$ . Apply Lemma 4.19 to  $f(t) = \Phi_c(5ut)$  with  $\theta = c/5$ , and sum over  $c$ . This gives

$$\log E_n(\omega_5^{-h}e^{-u}) = -\gamma(\sigma)/u + \Omega_h(\sigma) + R_h,$$

where  $\gamma(z) = [\text{Li}_2(e^{-z}) - \text{Li}_2(e^{-5z})]/5$ ;  $Q(s) = \sum_c \Phi_c(s) = \log(1 - e^{-5s}) - \log(1 - e^{-s})$ , independent of  $h$  because  $\prod_c(1 - \omega_5^{-hc}w) = (1 - w^5)/(1 - w)$ ;  $\Omega_h(\sigma) = \sum_c B_1(c/5)\Phi_c(\sigma)$ ; and

$$|R_h| \leq \bar{R} := (5W/3)|\varpi_\eta|\xi[1/(1 - \xi) + |\varpi_\eta|/(1 - \xi)^2].$$

*Derivation of  $\bar{R}$ .* Here  $f(t) = \Phi_c(5ut)$ , so  $f' = 5u\Phi'_c$  and  $f'' = 25u^2\Phi''_c$ , with  $|u| = W|\varpi_\eta|$  and  $\text{Re}(5ut) = 5Wt$ . From  $\Phi'_c(s) = \omega_5^{-hc}e^{-s}/(1 - \omega_5^{-hc}e^{-s})$ ,  $\Phi''_c(s) = -\omega_5^{-hc}e^{-s}/(1 - \omega_5^{-hc}e^{-s})^2$  and  $|1 - \omega_5^{-hc}e^{-s}| \geq 1 - e^{-\text{Re } s}$  we get  $|\Phi'_c(s)| \leq e^{-\text{Re } s}/(1 - e^{-\text{Re } s})$ , so  $|f'(n)| \leq 5W|\varpi_\eta|\xi/(1 - \xi)$  since  $\text{Re}(5un) = V$ ; and  $|\Phi''_c(s)| \leq e^{-\text{Re } s}/(1 - e^{-\text{Re } s})^2$ , so, as the denominator is at least  $(1 - \xi)^2$  for  $t \geq n$ ,

$$\int_n^\infty |f''| dt \leq \frac{25W^2|\varpi_\eta|^2}{(1 - \xi)^2} \int_n^\infty e^{-5Wt} dt = \frac{5W|\varpi_\eta|^2\xi}{(1 - \xi)^2}.$$

Lemma 4.19 gives  $\frac{1}{12}(|f'(n)| + \int |f''|)$  for each  $c$ ; summing over the four values of  $c$  gives  $\bar{R}$ .

The pairs  $c \leftrightarrow 5 - c$  have opposite  $B_1$  and conjugate roots of unity, so  $\text{Re } \Omega_h(W) = 0$ , i.e.  $|e^{\Omega_h(W)}| = 1$ . Also  $|\Omega_h| \leq \bar{\omega} := 0.8 \text{Li}_1(\xi)$ , since  $\sum_c |B_1(c/5)| = 0.8$ . (In `cls012_em_check.py` and `lap34_emchk.py` the bound exceeds the true remainder by a factor 5–20.)

4.5.3. *Classes 0, 1, 2. Saddle.* Define  $\alpha(V) = a_5 - \gamma(V) - VQ(V)/5$  and choose  $W$  by  $(m - 1/6)W^2 = \alpha(V)$ , where  $V = 5nW$ . With  $X = 1/W$  the exponent becomes

$$\frac{a_5}{u} + (m - \tfrac{1}{6})u - \frac{\gamma(\sigma)}{u} = X\Xi_V(\eta), \quad \Xi_V(\eta) = \frac{a_5 - \gamma(V\varpi_\eta)}{\varpi_\eta} + \alpha(V)\varpi_\eta.$$

$\Xi_V$  depends on  $V$  only. It satisfies  $\Xi'_V(0) = 0$  identically (this is the definition of  $\alpha$ ), and  $\Xi(-\eta) = \overline{\Xi(\eta)}$ . Put  $\Xi_2 = -\Xi''(0) > 0$  and  $\text{Amp} = W^{3/2}e^{X\Xi(0)}/\sqrt{2\pi\Xi_2}$ . Then

$$c(m) = \text{Amp} \left[ M + \sum_h A_h \omega_5^{hm} e^{\Omega_h} \delta_h \right] + E'_2, \quad M = \sum_h A_h \omega_5^{h \cdot \text{cls}} e^{\Omega_h(W)} \in \mathbb{R}.$$

As  $V \rightarrow \infty$ ,  $M \rightarrow \mathfrak{a}(\text{cls})$ .  $M$  is the main term with the correct normalisation  $|E_n(\omega_5 e^{-W})|$ .

*Error terms.* All are relative to  $\text{Amp}$ , and each is either independent of  $X$  or non-increasing in  $X$ . Set  $\eta_0 = c'_c/\sqrt{X\Xi_2}$  with  $c'_c = 4$ , and  $s = \eta\sqrt{X\Xi_2}$ .

- *Window*,  $|\eta| \leq \eta_0$ . Write  $\psi = X(ic_3\eta^3 + R_4) + \Delta\Omega + R_h$ . The cubic term is a pure phase, so  $|e^\psi - 1| \leq \min(k_3 s^3, 2)e^r + e^r - 1$  with  $r = k_4 s^4 + k_1 s + \bar{R}$ , where  $k_3 = |c_3|/(\Xi_2^{3/2}\sqrt{X})$ ,  $k_4 = \sup |\Xi''''|/(24\Xi_2^2 X)$  and  $k_1 = \beta_1/\sqrt{X\Xi_2}$ . Here  $\beta_1 = 0.8V\xi/(1 - \xi)$  bounds  $|d\Omega/d\eta|$ . The supremum of  $|\Xi''''|$  is enclosed on sub-balls of width 0.01 covering  $[0, \eta_0(X_{\min})]$ . The Gaussian truncation contributes  $\text{erfc}(c'_c/\sqrt{2})$ .
- *Mid region*,  $\eta_0 < |\eta| \leq 2$ .  $D(\eta) = \text{Re}(\Xi(0) - \Xi(\eta)) \geq d_k \eta^2$  on 170 fixed  $\eta$ -tiles, using the better of the Taylor bound  $\Xi_2/2 - \sup |\Xi''''|\eta^2/24$  and a direct ball evaluation. The contribution is  $e^{2\bar{\omega} + \bar{R}} \sum_k \sqrt{\Xi_2/(2d_k)} \text{erfc}(\max(c'_c \sqrt{d_k/\Xi_2}, \eta_k \sqrt{X_{\min} d_k}))$ .
- *Crude terms*  $E_2$ . These are: the  $(1 + a_5/u)$  part of  $r_1$ ;  $|y| \in [2W, 1]$ , with  $|r_1| \leq e^{a_5 X/(1+\eta_1^2)} + 1 + a_5 X/\eta_1$ ;  $|y| > 1$ , with  $|r_1| \leq (a_5^2/2y^2)e^{a_5}$ ;  $e(m) \leq e^{mW} E_n(e^{-W})$ ;  $a_5|H_h| \leq a_5 e^{(m-1)W} E_n(e^{-W})$ ; and the  $\rho$ -term, by Rankin's trick at the same  $W$ , with  $\mathcal{P}_k(i) \leq W(e^{a_k X} - 1)e^{(i-1/6)W}$  and  $k \leq K(m)$ .

*The  $\rho$ -term in detail.* By §4.5.1,  $|\rho| \leq \sum_{j < m} e(j)R(m-j)$ , where  $R(i) = \sum_{5|k, 10 \leq k \leq K(i)} \varphi(k)\mathcal{P}_k(i) + E_{\text{con}}$ . Every  $i = m - j$  here is  $\geq 1$ , so Theorem 4.3 applies. Write  $x = i - \frac{1}{6}$  and  $a_k = 2\pi^2/(3k^2)$ , so that  $a_{10} = a_5/4$  and  $a_k \leq a_5/9$  for  $k \geq 15$ .

- *Bessel part.* The series of  $I_1$  gives  $\mathcal{P}_k(i) = \sum_{r \geq 1} a_k^r x^{r-1}/(r!(r-1)!)$ . With  $x^{r-1}/(r-1)! \leq e^{xW}/W^{r-1}$  this gives  $\mathcal{P}_k(i) \leq W(e^{a_k X} - 1)e^{(i-1/6)W}$ .
- *Counting.*  $\varphi(10) = 4$ , and  $\sum_{j=3}^{K/5} \varphi(5j) \leq \sum 4j \leq 2\bar{K}(\bar{K}+1)$ , where  $\bar{K} = K/5$  and  $K$  bounds  $K(i)$  for every  $i \leq m$ . The code uses  $K = \sqrt{4\pi(a_5 X^2 + 1)} + 2$ , since  $m - \frac{1}{6} = \alpha X^2 \leq a_5 X^2$ .
- *Rankin.* Since  $e(j) \geq 0$ ,  $\sum_{j < m} e(j)e^{(m-j-1/6)W} \leq e^{(m-1/6)W} E_n(e^{-W})$ . So the Bessel part contributes at most  $W[4e^{a_5 X/4} + 2\bar{K}(\bar{K}+1)e^{a_5 X/9}]e^{(m-1/6)W} E_n(e^{-W})$ .
- *$E_{\text{con}}$  part.*  $e^{(m-j)W} \geq 1$  for  $j \leq m$ , so  $\sum_{j < m} e(j)E_{\text{con}} \leq E_{\text{con}}e^{mW} E_n(e^{-W}) = E_{\text{con}}e^{W/6} \cdot e^{(m-1/6)W} E_n(e^{-W})$ .

Together these give the term  $T_\rho = F_m e^{W/6} (W[4e^{a_5 X/4} + e^{a_5 X/9} 2\bar{K}(\bar{K}+1)] + E_{\text{con}})$ , where  $F_m$  is an upper bound for  $e^{(m-1/6)W} E_n(e^{-W})/\text{Amp}$ . The script also applies the factor  $e^{W/6}$  to the Bessel part, where it is not needed; this only enlarges the bound.

All the crude terms use  $E_n(e^{-W}) \leq \exp(0.8X \text{Li}_2(\xi) + 4\text{Li}_1(\xi))$ , which holds because  $E_n$  has non-negative coefficients, so  $|E_n(\omega_5^{-h} e^{-u})| \leq E_n(e^{-W})$ . The total is at most  $\sqrt{2\pi\Xi_2} X^{3/2} e^{-cX} \text{poly}(X)$ , with  $c \geq \kappa - a_5/4$  and  $\kappa = a_5 - \gamma - 0.8\text{Li}_2(\xi)$ ; the rate  $a_5/4$  comes from the  $k = 10$  Rankin term. The script asserts  $c > 0$  and  $X_{\min} > 4.5/c$ , and  $E_2 \leq 2.3 \cdot 10^{-10}$ .

*Certified condition.*  $\text{sgn}(\mathbf{a})M - 4\epsilon_{\text{tot}} - E_2 > 0$ , where  $\epsilon_{\text{tot}} = \epsilon_{\text{win}} + \text{erfc}(c'/\sqrt{2}) + \epsilon_{\text{mid}}$  (called `delta` in the log). Each factor  $A_h \omega_5^{hm} e^{\Omega_h(W)}$  is unimodular, because  $\text{Re } \Omega_h(W) = 0$ , and  $|\delta_h| \leq \epsilon_{\text{tot}}$ . So the bracket is  $M$  plus at most  $4\epsilon_{\text{tot}}$ , and the condition implies that  $c(m)$  has the sign of  $\mathbf{a}(\text{cls})$ .

*Range of  $X$ .*  $X = 5n/V \geq 1455/V$  for  $n \geq 291$ .  $\lambda \geq 4$  gives  $\alpha X^2 \geq 4VX + 23/6$ , so  $X \geq 4V/a_5$ . Per tile,  $X_{\min} = \max(1455/V_{hi}, 4V_{lo}/a_5)$ .

*Monotonicity in  $X$ .*  $k_3, k_1 \propto X^{-1/2}$  and  $k_4 \propto X^{-1}$ ;  $\bar{R} \propto W$ ; the  $\text{erfc}$  arguments are non-decreasing in  $X$ ; the window sets are taken at  $\eta_0(X_{\min})$ ; the crude terms decay. So one evaluation at  $X_{\min}$  per tile covers every  $X \geq X_{\min}$ , that is, every  $(n, m)$  whose saddle lies in the tile. No monotonicity in  $\lambda$  is used.

**Lemma 4.20** (classes 0–2, tiles). *On 500 contiguous tiles of width 0.02 covering  $V \in [2, 12]$  (the last tile snapped to  $[11.98, 12]$ ), the certified condition holds, with 0 failures. The minimum margins  $\text{sgn}(\mathbf{a})M - 4\epsilon_{\text{tot}} - E_2$  are 3.0700 (class 0,  $|\mathbf{a}| = 3.618$ ), 1.6879 (class 1,  $|\mathbf{a}| = 2.236$ ) and 0.8339 (class 2,  $|\mathbf{a}| = 1.382$ ). The worst  $\epsilon_{\text{tot}}$  is 0.137, at  $V = 9.78$ ,  $X_{\min} = 148.6$ .*

Certificate: `cls012_lap.py 2.0 12.0 0.02; log third/logs/cls012_lap_rerun.log (line 1)`.

*Tail cell  $V \geq 11.99$ .* Write  $\Xi = a_5/\varpi_\eta + a_5\varpi_\eta + \Delta$ . On the strip  $|\text{Im } \eta| \leq 1/2$  we have  $\text{Re } \varpi_\eta \geq 1/2$ , so  $|\gamma(V\varpi_\eta)/\varpi_\eta| \leq \frac{12}{25} \text{Li}_2(e^{-V/2})$ . Cauchy estimates on discs of radius  $1/2$  give  $\Xi_2 \in 2a_5 \pm 8\epsilon$ ,  $|c_3| \leq a_5 + 8\epsilon$  and  $|\Xi''''| \leq 24a_5 + 384\epsilon$ . The mid-region bound is  $a_5\eta^2/(1+\eta^2) - \frac{12}{25} \text{Li}_2$ . Also  $M \geq |\mathbf{a}| - 4(e^{\bar{\omega}} - 1)$  and  $X \geq 4V/a_5 \geq 4V_T/a_5$ . Every  $V$ -dependent quantity moves in the safe direction as  $V$  grows, so a single evaluation at  $V_T$  covers all  $V \geq V_T$ .

**Lemma 4.21** (classes 0–2, tail cell). *For all  $V \geq V_T = 11.99$  the certified condition holds, with margins 3.0950, 1.7130, 0.8589 (classes 0, 1, 2) and  $X_{\min} = 182.2$ . The cell overlaps the last tile on  $[11.99, 12]$ , so coverage at  $V = 12$  does not depend on the snap.*

Certificate: `cls012_lap.py tail 11.99; log third/logs/cls012_lap_rerun.log (line 2)`.

*Validation (not part of the proof).* `cls012_asym_check.py:  $c/(\text{Amp} \cdot M) = 0.990\text{--}0.998$  at  $n = 60, 100$ . An audit checked 45 exact points,  $n \in \{291, 437, 500, 700, 1500, 2500\}$ ,  $V = 2.08\text{--}40$ ,`

including  $\lambda \approx 4$  and  $\mu \approx 0.105$ : every sign correct;  $|c/\text{Amp} - M|$  ranged over 0.0005–0.0185, against a certified bound of 0.27–0.54.

**4.5.4. Classes 3 and 4: the re-centred (thin) saddle.** For classes 3 and 4,  $\sum_h A_h \omega_5^{h \cdot \text{cls}} = \mathbf{a}(\text{cls}) = 0$ , so the constant term of the twisted sum cancels. The survivor carries  $e^{-5nu} = e^{-\sigma}$ , which is an  $O(N)$  term in the exponent. It must not be treated as an amplitude (see Appendix A.3); here it is moved into the exponent.

*The thin function.* Put

$$g(s) = e^s \sum_{h=1}^4 A_h \omega_5^{h \cdot \text{cls}} \exp(\Omega_h(s)), \quad s = \sigma = V\varpi_\eta.$$

Since  $\omega_5^{hm} = \omega_5^{h \cdot \text{cls}}$ , we have  $\sum_h A_h \omega_5^{hm} e^{\Omega_h(\sigma)} = e^{-\sigma} g(\sigma)$  exactly.

- Lemma 4.22** (the thin function). (1) Leading coefficient. Expand  $e^{\Omega_h} = 1 + \Omega_h + \dots$ . The constant term cancels. The coefficient of  $e^{-s}$  is  $f_\infty = -\sum_h A_h \omega_5^{h \cdot \text{cls}} \sum_c B_1(c/5) \omega_5^{-hc} = -1$  exactly, for both classes (checked in Arb to  $10^{-37}$ , `lap34-lib.finf`).
- (2) Uniform bound. For every  $s$  with  $\text{Re } s = V$ ,  $|g(s) - f_\infty| \leq \varepsilon_g(V) := 4e^V [0.8(\text{Li}_1(\xi) - \xi) + \bar{\omega}^2 e^{\bar{\omega}}/2]$ . This follows from  $|\Omega_h - \Omega_h^{(1)}| \leq 0.8 \sum_{r \geq 2} \xi^r/r$  and  $|e^\Omega - 1 - \Omega| \leq |\Omega|^2 e^{|\Omega|}/2$ , where  $\Omega_h^{(1)}$  is the first-order term.  $\varepsilon_g$  is  $\xi^{-1}$  times a series in  $\xi$  with non-negative coefficients starting at  $\xi^2$ , so it decreases in  $V$ . For example  $\varepsilon_g(11.99) = 1.787 \cdot 10^{-5}$ .
- (3) Reality.  $g(V)$  is real for real  $V$ , because  $h \leftrightarrow 5-h$  are conjugate. The Arb enclosures of  $g$  on the  $V$ -tiles lie in  $[-1.13, -0.96]$ .

(An audit tested (2) at 6000 points and found  $\max |g + 1|/\varepsilon_g = 0.233$ .)

*Saddle.* Choose  $W$  by  $\alpha(V)X^2 = m - \frac{1}{6} - 5n$ . The exponent  $a_5/u - \gamma/u + (m - 1/6)u - 5nu$  then equals  $X\Xi_V(\eta)$ , with the same  $\Xi_V$  as in §4.5.3. So

$$c(m) = \text{Amp}[g(V) + \Delta], \quad \text{Amp} = W^{3/2} e^{X\Xi_V(0)} / \sqrt{2\pi\Xi_2} > 0,$$

and the certificate proves  $|\Delta| < -g(V)$ , which gives  $c(m) < 0$ . All the  $O(N)$  parts of  $\log F_m$  are inside  $X\Xi_V$ : the kernel, the twisted-product term  $-\gamma/u$ , and the thin shift  $-5nu$ . The remaining  $\log g(V\varpi_\eta)$  is  $O(1)$ . It is kept out of the Gaussian and paid for in the window error through  $\sup |g'|$ .

*Error terms* (relative to Amp, all non-increasing in  $X$ ).

- *Window.* With  $G(\eta) = e^{V\varpi_\eta} \sum_h A_h \omega_5^{hm} e^{\Omega_h + R_h}$ ,  $\psi = X(c_3\eta^3 + R_4)$  and  $c_3$  purely imaginary,

$$|e^\psi G - g(V)| \leq e^{k_4 s^4} (V|\eta| \sup_{\text{win}} |g'| + \varepsilon_R) + |g(V)| (\min(k_3 s^3, 2) e^{k_4 s^4} + e^{k_4 s^4} - 1).$$

Here  $\varepsilon_R = 4e^{\bar{\omega}} e^V (e^{\bar{R}} - 1) \geq |G - g(V\varpi_\eta)|$ ;  $\sup |g'|$  is the smaller of a ball evaluation (sub-balls of width at most 0.05 in  $\text{Im } s$ ) and the Cauchy bound  $\varepsilon_g(V - 1)$ ; the integral is an upper Riemann sum over 800  $s$ -cells; the Gaussian truncation contributes  $|g| \text{erfc}(c'_c/\sqrt{2})$ .

- *Mid region*,  $\eta_0 < |\eta| \leq 2$ . The bound  $\text{Re}(\Xi(0) - \Xi(\eta)) \geq d_k \eta^2$  on 170  $\eta$ -tiles. On each tile,  $\sup |G| \leq \sup |g(V\varpi_\eta)| + \varepsilon_R$ . Tiles whose  $\text{erfc}$  argument exceeds 40 are bounded jointly, via  $4e^{V+\bar{\omega}+\bar{R}} \cdot 170 \sqrt{\Xi_2/2d_{\min}} \text{erfc}(40)$ .
- *Crude terms.* As in §4.5.3, but with an extra factor  $e^V$ , because Amp carries  $e^{-V}$ . The rate is  $\kappa = a_5 - \gamma - 0.8 \text{Li}_2(\xi)$ . Tiles use  $e^{V_{hi}}$ ; the tail cell uses  $V \leq a_5 X/3$ . The script asserts decay  $c \geq \kappa - \max(a_5/5, a_5/4)$  (tiles), or  $c \geq \kappa - a_5/3 - a_5/4$  (tail), and  $X_{\min} > 4.5/c$ .  $E_2 \leq 10^{-8}$  everywhere.

*Range of  $X$ .*  $X = 5n/V \geq 4105/V$  for  $n \geq 821$ .  $\lambda \geq 4$  gives  $m - 1/6 - 5n \geq 15n + 23/6 > 3VX$ , so  $X > 3V/a_5$ . Per tile,  $X_{\min} = \max(4105/V_{hi}, 3V_{lo}/a_5)$ . For the tail cell,  $X \geq X^* = \sqrt{15 \cdot 821/a_5} = 216.3$ .

**Lemma 4.23** (classes 3–4). (1) On 500 tiles of width 0.02 covering  $V \in [2, 12]$  (last tile snapped to  $[11.98, 12]$ ),  $|\Delta| < -g(V)$  with 0 failures; the minimum margins  $-g - |\Delta|$  are 0.8363 (class 3) and 0.8361 (class 4), on the tile  $[11.98, 12]$  (window 0.119, mid 0.005,  $\varepsilon_R$  0.042).  
 (2) On the analytic tail cell  $V \geq V_T = 11.99$ , the margin is 0.8146 for both classes (0.81456). The tail cell uses the strip bounds of §4.5.3 together with  $|g(V\varpi_\eta) - g(V)| \leq 2\varepsilon_g(V_T)$  and  $|g(V)| \geq 1 - \varepsilon_g(V_T)$ , where  $\varepsilon_g(11.99) = 1.787 \cdot 10^{-5}$ ; since  $\varepsilon_g$  decreases, these bounds hold for every  $V \geq 11.99$ . The tail cell overlaps the last tile on  $[11.99, 12]$ .

The certified relative error is therefore at most 0.19.

Certificate: `lap34_cert.py 2.0 12.0 0.02; lap34_cert.py tail 11.99; log third/logs/lap34_cert.log`.

Validation (not part of the proof). `lap34_validate.py`: 70 exact points at  $n = 150, 300, 821, 2000$ , with  $\mu$  up to 0.1050 and  $\lambda$  down to 4.0; the true  $|c/\text{Amp} - g|$  lies in  $1.8 \cdot 10^{-4}$ – $1.6 \cdot 10^{-2}$ , which is 24–435 $\times$  below the certified bound, and every sign is negative. Two audits checked a further 36 and 31 points with independent exact code (the second by a multi-modular CRT method that shares no code with the others), up to  $n = 5000$  and  $V = 46.8$ , including points straddling the tile/tail junction: every sign negative, true error 28–579 $\times$  below the certified bound.

4.5.5. *Coverage lemma for the saddle.* Write  $A(V) := \alpha(V)/V^2$ . Then  $A$  is continuous,  $A \rightarrow 1/5$  as  $V \rightarrow 0$ , and  $A \rightarrow 0$  as  $V \rightarrow \infty$ ; and  $\alpha$  is increasing, since  $\alpha' = Vq(V)/5$  with  $q = 1/(e^V - 1) - 5/(e^{5V} - 1) > 0$ .

**Lemma 4.24.**  $A \geq 0.04424$  on  $[0.001, 2]$  (5858 tiles, using  $A \geq \alpha(v_1)/v_2^2$ ) and  $A \geq 0.1998$  on  $(0, 0.001]$ .

Certificate: `cls012_cov.py; log third/logs/cls012_cov.log`.

For classes 0–2 the saddle equation is  $A(V) = (m - 1/6)/(25n^2) < 0.4\mu$ ; for classes 3–4 it is  $A(V) = (m - 1/6 - 5n)/(25n^2)$ , which is smaller still. When  $\mu < 0.105$ , the right-hand side is  $< 0.042$ . By the intermediate value theorem a root exists, and every root has  $V > 2$ . The representation is exact for every  $W > 0$ , so any root may be used.

4.6. **The centre**,  $\mu \geq 0.105$ . Throughout this subsection  $n \geq 821$  and  $0.105 \leq \mu \leq \frac{1}{2}$ . We use  $r = e^{-L/(5n)}$  and  $q = \omega_5^h e^{-\sigma/(5n)}$ , with  $\sigma = L - iy$ , so that  $y = 5n(\theta - 2\pi h/5)$ ;  $\Phi_c(s) = \log(1 - \omega_5^{hc} e^{-s})$  (principal branch) and  $Q = \sum_{c=1}^4 \Phi_c = \log \frac{1 - e^{-5s}}{1 - e^{-s}}$ ;  $\Lambda(\sigma) = \int_0^1 Q(t\sigma) dt$ ; and  $C_h(\sigma) = \sum_c (c/5 - \frac{1}{2})(\Phi_c(\sigma) - \Phi_c(0))$ . The starting point is Cauchy’s formula for the polynomial  $P_n$ :

$$c(m) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_n(re^{i\theta}) r^{-m} e^{-im\theta} d\theta, \quad r^{-m} = e^{2n\mu L}.$$

4.6.1. *Near-peak piece* ( $A$ ):  $|y| \leq 1.1$  around  $\theta = 2\pi h/5$ ,  $h = 1, \dots, 4$ .

**Lemma 4.25** (finite Euler–Maclaurin). For  $0 \leq a < 1$ ,

$$\sum_{t=0}^{n-1} f(t+a) = \int_0^n f + B_1(a)(f(n) - f(0)) + \frac{B_2(a)}{2}(f'(n) - f'(0)) - \int_0^n \frac{\tilde{B}_2(x-a)}{2} f''(x) dx.$$

*Proof.* Apply the Euler–Maclaurin formula of order 2 with offset  $a$  on each  $[t, t+1]$  and sum over  $t$ ; the argument is the same as for Lemma 4.19, but on a finite range.  $\square$

The constants are  $B_2(1/5) = B_2(4/5) = 1/150$  and  $B_2(2/5) = B_2(3/5) = -11/150$ . Since  $\tilde{B}_2 \in [-1/12, 1/6]$ , the last term is at most  $\frac{1}{12} \int |f''|$ . The identity was checked numerically to  $10^{-15}$ .

**Lemma 4.26.** Apply Lemma 4.25 to  $f(x) = \Phi_c(x\sigma/n)$  with  $a = c/5$ , and sum over  $c$ . The terms  $k = 5t + c$ ,  $t < n$ , are exactly the factors of  $P_n$ . This gives

$$|\log P_n(q) - n\Lambda(\sigma) - C_h(\sigma)| \leq \frac{K_h(\sigma)}{n},$$

where

$$K_h = \sum_c \left[ \frac{|B_2(c/5)|}{2} |\sigma| (|\Phi'_c(\sigma)| + |\Phi'_c(0)|) + \frac{|\sigma|^2}{12} \sup_{t \in [0,1]} |\Phi''_c(t\sigma)| \right],$$

valid for  $|y| < 2\pi/5$  and  $L \geq 0$ , uniformly down to  $L = 0$ . The reason is that  $\Phi_c$  has no singularity on the segment  $[0, \sigma]$ , because  $\omega_5^{hc} \neq 1$ .

Two exact facts are used:  $|e^{C_h(L)}| = 1$  for real  $L$  (enclosed in the certificate, not assumed), and the four peaks have equal modulus exactly. The class margin is then

$$M_s(L) = \text{sgn}_s \text{Re} \sum_h \omega_5^{-hs} e^{C_h(L)}, \quad \text{sgn}_0 = +1, \text{sgn}_{1,\dots,4} = -1.$$

*Saddle.*  $\mu = -\Lambda'(L)/2$ . Since  $Q'(0) = \sum_c \omega_5^c / (1 - \omega_5^c) = -2$ , we get  $\Lambda'(0) = -1$  and  $\mu(0) = 1/2$ .

**Lemma 4.27** (saddle endpoint).  $\mu(2.1) < 0.105$ , by the real mean-value theorem on 4096  $t$ -pieces: the log prints `mu(2.1) = -p'/2 in [0.104 +/- 3.02e-4] < 0.105: True`.

Certificate: `centre3_saddle_end.py`; log `third/logs/centre3_saddle_end.log`.

(An audit computes  $\mu(2.1) = 0.103772$  from the closed form, so  $L(0.105) = 2.0814$ .) The near-peak certificate asserts  $a_L := \Lambda''(L) > 0$  on every  $L$ -box, so  $\mu(L)$  is strictly decreasing. Therefore every  $\mu \in [0.105, 1/2]$  has a unique saddle  $L \in [0, 2.1]$ .

*Laplace step.* Put

$$\mathcal{N}(n, L) = \frac{e^{n\Lambda(L) + 2n\mu L} \sqrt{2\pi/(na_L)}}{10\pi n}.$$

Then

$$c(m) = \mathcal{N}(n, L) \left[ \sum_{h=1}^4 \omega_5^{-hm} e^{C_h(L)} (1 + \rho_h) + \text{strip} + \text{OFF} \right],$$

using  $d\theta = dy/(5n)$ ; the linear phase cancels at the saddle, since  $-in\Lambda'(L)y = 2in\mu y$ ; and each peak contributes  $\mathcal{N}\omega_5^{-hm} e^{C_h}$ . The three error pieces are bounded as follows.

- *Inner window*  $|y| \leq 0.35$ . Write  $\psi(y) = \Lambda(L - iy) - \Lambda(L) + i\Lambda'(L)y$  and  $Z = n\psi + na_L y^2/2 + \Delta C$ . The odd part  $T = n\psi_3 y^3 + C_1 y$  integrates to exactly 0 against the Gaussian. The rest is bounded using  $|e^Z - 1 - Z| \leq \frac{|Z|^2}{2} \max(1, e^{\text{Re } Z})$ ;  $\text{Re } \psi \leq -by^2/2$ , where  $b$  is a certified lower bound for  $\text{Re } \Lambda''$  on the window; Taylor constants from  $\Lambda^{(k)}(\sigma) = \int_0^1 t^k Q^{(k)}(t\sigma) dt$ , enclosed on  $t$ -pieces; the factor  $e^{K/n} - 1$  and the Gaussian tail.
- *Strip*  $0.35 \leq |y| \leq 1.1$ . Lemma 4.26 is combined with  $\text{Re } \psi(y) = -\int_0^{|y|} (|y| - u) \text{Re } \Lambda''(L - iu) du$ , with  $\text{Re } \Lambda''$  bounded below on boxes. This gives terms of the form  $\sqrt{n} e^{-nD}$ , which are non-increasing for  $n \geq 1/(2D)$ ; that condition is asserted. The strip and the Gaussian tail are normalised with the upper curvature  $a_{up}$ , which is the safe direction.
- *Uniformity in  $n$* .  $n$  enters only through  $K/n$ , the  $1/n$  moment terms, the Gaussian tail  $e^{-nAY^2/2}/\sqrt{n}$ , and the strip. Each of these is non-increasing in  $n$ . So a bound at  $n = N_0 = 821$  holds for all  $n \geq 821$ .

**Lemma 4.28** (near peak). On 210  $L$ -boxes of width 0.01 covering  $[0, 2.1]$ , each with 28 inner  $y$ -boxes, 60 strip  $y$ -boxes and 64  $t$ -pieces, every quantity enclosed over its full box, at  $N_0 = 821$ : for every box and every class,  $M_s > \sum_h |e^{C_h}|(\rho_h + \text{strip}_h) =: \text{err}_s$ . The worst ratios  $\text{err}/\text{margin}$  are

0.0566, 0.2225, 0.2241, 0.7977, 0.8469 for the classes 0, 1, 2, 3, 4. The minimum slack  $\min_s(M_{s,lo} - \text{err}_{s,up})$  is 0.018703, attained for class 4 on the box  $L \in [2.09, 2.10]$ .

Certificate: `centre3_peak.py 821 210 64` (reproduced inside `centre3_comb.py`); `log third/logs/centre3_peak_821.log, centre3_comb_821.log`.

Since  $L(0.105) = 2.0814$ , only the boxes up to  $[2.08, 2.09]$  are needed; the last box  $[2.09, 2.10]$  is certified as well.

#### 4.6.2. The off-peak lemma (B).

**Lemma 4.29** (off-peak lemma). *Let  $n \geq 821$  and  $L \in [0, 2.1]$ , and let  $\theta$  satisfy  $|5n(\theta - 2\pi h/5)| > 1.1$  for  $h = 1, \dots, 4$ , taken mod  $2\pi$ . Then*

$$\log |P_n(e^{-L/(5n)} e^{i\theta})| - n\Lambda(L) \leq -\mathcal{E}(n), \quad \mathcal{E}(n) = 1.5 \log n + 6.3 \quad (\mathcal{E}(821) = 16.36).$$

Every certificate uses the constant 6.3 as the exact ball `arb('6.3')`. The proof occupies the rest of this subsection: the  $\theta$ -circle is covered by the pieces (E), (E0), (N) and (F) below, each certified.

*Closure.* For fixed  $n$ ,  $\log |P_n(e^{-L/(5n)} e^{i\theta})|$  is continuous in  $(\theta, L)$ , with values in  $[-\infty, \infty)$ . So each bound “ $\leq n\Lambda(L) - \mathcal{E}(n)$ ” certified on a box also holds on the closure of that box. This closes the float-endpoint slivers  $|y| \in (1.1, \text{fl}(1.1))$  at the start of (E) and  $(\text{fl}(\pi), \pi]$  at the end of (F) (the Arb run of (F) also closes the latter by ending at `nextafter(pi, 4)`). The sliver  $L \in [0.1, \text{fl}(0.1))$  at the lower end of the  $L$ -boxes of (E), (E0) and (N), where  $\text{fl}(0.1) = 0.1 + 5.5 \cdot 10^{-18}$ , is closed directly: the lowest  $L$ -box of these pieces starts at 0.0999, below the exact 0.1, so  $L = 0.1$  is inside a certified box and the reduction of Lemma 4.30 from  $L < 0.1$  to  $L' = 0.1$  costs exactly  $n(0.1 - L) \leq 0.1n$ , which is what each piece charges (`arb('0.1')`).

Since  $P_n$  has real coefficients,  $|P_n(\bar{q})| = |P_n(q)|$ , so it suffices to take  $\theta \in [0, \pi]$ . The pieces (E) and (N) use both signs of  $y$  near  $h = 1, 2$ , which also covers  $h = 3, 4$ .

*Closed forms.* For  $h \neq 0$ ,  $\Lambda(\sigma) = (\text{Li}_2(e^{-5\sigma})/5 - \text{Li}_2(e^{-\sigma}) + 4\zeta(2)/5)/\sigma$ , using  $\sum_{c=0}^4 \text{Li}_2(\omega_5^c x) = \text{Li}_2(x^5)/5$ . For  $h = 0$ ,  $\Lambda_0(\sigma) = 4(\text{Li}_2(e^{-\sigma}) - \zeta(2))/\sigma$ .  $Q$  is decreasing and convex with  $|Q'| \leq 2$ : indeed  $Q'' = [g_2(s/2) - g_2(5s/2)]/s^2$  with  $g_2(x) = x^2/\sinh^2 x$  decreasing. Since  $\Lambda' = (Q - \Lambda)/L$ ,  $\Lambda$  is decreasing on  $L > 0$ .

**Lemma 4.30** (monotonicity in  $L$ ). *For  $0 \leq L \leq L'$ ,  $\log |P_n(re^{i\theta})| \leq \log |P_n(r'e^{i\theta})| + n(L' - L)$ , where  $r = e^{-L/(5n)}$  and  $r' = e^{-L'/(5n)}$ .*

*Proof.* We have  $|1 - \varrho e^{ia}|^2 = (1 - \varrho)^2 + 4\varrho \sin^2(a/2)$ . For  $\varrho' \leq \varrho \leq 1$ , this gives  $|1 - \varrho e^{ia}|^2 \leq (\varrho/\varrho')|1 - \varrho' e^{ia}|^2$ . Take  $\varrho = r^k$ ,  $\varrho' = r'^k$ , and use  $\sum_{k \leq 5n, 5 \nmid k} k = 10n^2$ : the total factor on  $\log |P_n|$  is  $\frac{1}{2} \cdot \frac{L' - L}{5n} \cdot 10n^2 = n(L' - L)$ .  $\square$

Pieces (E), (E0) and (N) are certified on  $L \in [0.1, 2.1]$ . For  $L < 0.1$  they use Lemma 4.30 at  $L' = 0.1$  together with  $n\Lambda(L) \geq n\Lambda(0.1)$  ( $\Lambda$  decreasing, and  $y$  unchanged because  $\theta$  is fixed). The cost is  $0.1n$ , which the boxes touching  $L = 0.1$  are required to absorb.

Piece (E):  $h \in \{1, 2\}$ ,  $1.1 \leq |y| \leq 6$ .  $\Phi_c$  is analytic on  $\text{Re } s > 0$  and at  $s = 0$ . So Lemma 4.25 applies for every  $y$ , even past  $|\sigma| = 2\pi/5$ ; the disc was needed only for the Taylor step in §4.6.1. This gives  $\log |P_n| \leq n \text{Re } \Lambda(\sigma) + \text{Re } C_h(\sigma) + K_h(\sigma)/n$ , with the remainder in integral form. The certificate uses 20000 adaptive boxes. On each box it checks  $\mathcal{V}(N_0) = -N_0 g_E + \text{Re } C + K/N_0 + \mathcal{E}(N_0) < 0$ , where  $g_E = \Lambda(L_{hi}) - \text{Re } \Lambda(\sigma) - 0.1\chi$ , and  $g_E > 1.5/N_0$ , so that  $\mathcal{V}(n)$  is non-increasing in  $n$ . The largest  $K$  is  $5.6 \cdot 10^3$ , near a singular crossing at  $L = 0.1$ . The logged worst values of  $\mathcal{V}(N_0)$  are  $-2.49, -2.27$  ( $h = 1, y \gtrless 0$ ) and  $-2.55, -2.18$  ( $h = 2$ ).

Certificate: `offpeak_em.py 821 6.0 32`; `log third/logs/offpeak_em_821.log`.

Piece (E0):  $h = 0$ ,  $|y| \leq 20$ . Here  $\Phi = \log(1 - e^{-s})$  is singular at 0, so the  $t = 0$  terms are split off and Lemma 4.25 is applied on  $t = 1, \dots, n-1$ ; note  $\sum_c B_1(c/5) = 0$ . This gives

$$\log |P_n| \leq n \operatorname{Re} \Lambda_0(\sigma) + \Psi_0(\varsigma) + 0.08(|\varsigma \Phi'(\varsigma)| + |\varsigma| |\Phi'(\sigma)|) + \frac{|\varsigma| |\sigma|}{3} \int_{1/n}^1 |\Phi''(t\sigma)| dt,$$

where  $\varsigma = \sigma/n$ ,  $\Psi_0(\varsigma) = 4 + \log(24/625) + \sum_c \operatorname{Re} \ell(c\varsigma/5) - 4 \operatorname{Re}[\varsigma^{-1} \int_0^\varsigma \ell]$  and  $\ell(s) = \log((1 - e^{-s})/s)$ . The  $\log |\varsigma|$  terms cancel exactly. For  $\operatorname{Re} s \geq 0$  and  $|s| \leq 0.1$ :  $|(1 - e^{-s})/s - 1| \leq |s|e^{|s|}/2$ , hence  $|\ell(s)| \leq |s|$ ;  $|s\Phi'(s)| \leq 1/(1 - |s|e^{|s|}/2)$ ; and  $|\Phi''(s)| \leq 1/(|s|^2 m_0^2)$ , with  $m_0 = 1 - |s|e^{|s|}/2$ . Consequently  $\Psi_0 \leq 0.7403 + 6|\sigma|/N_0$ . The certificate uses 4945 boxes, and the worst value is  $\mathcal{V} = -0.23$ ; this is a bisection artefact, as the true slack is in the hundreds.

*Certificate:* `offpeak_em0.py 821 20 32; log third/logs/offpeak_em0_821.log`.

Piece (N):  $6 \leq |y| \leq 0.25n$  ( $20 \leq y$  for  $h = 0$ ). Since  $|q| < 1$ ,

$$\log |P_n| = -\operatorname{Re} \sum_j G_j/j \leq \sum_j |G_j|/j, \quad G_j = \frac{(1 - e^{-j\sigma})A_j}{1 - e^{-j\varsigma}}, \quad A_j = \sum_c \omega_5^{hcj} e^{-cj\varsigma/5},$$

with  $\varsigma = \sigma/n$ ;  $A_j$  is the sum over the four  $c$  for fixed  $j$  (the factor  $1 - e^{-j\sigma}$  comes from summing the geometric series over  $t < n$ ). The bounds on  $A_j$  are  $|A_j| \leq \min(4, \bar{A}_j + 2j|\varsigma|)$  and  $|A_j| \leq \alpha_N(jL/n)$  with  $\alpha_N(a) = \sum_c e^{-ca/5}$ , where  $\bar{A}_j = |\sum_c \omega_5^{hcj}|$  is 1 or 4.

- *Part I*,  $j \leq \pi n/y$ . Use  $1/|1 - e^{-z}| \leq 1/|z| + c_1$  with  $c_1 = 1 + 1/\pi$ . This is the maximum modulus of  $1/(1 - e^{-z}) - 1/z$  on the rectangle  $[0, 20] \times [-\pi, \pi]$ : it is at most 0.593 on the left edge,  $1 + 1/\pi$  on the top and bottom edges, and 1.06 on the right edge. Every  $z = j\varsigma$  in Part I lies in the rectangle, since  $|\operatorname{Im} z| = jy/n \leq \pi$  and  $\operatorname{Re} z = jL/n \leq \pi L/y \leq 1.1$ . This gives Part I  $\leq nM_N(\sigma) + R_1$ , where  $M_N(\sigma) = \sum_j |1 - e^{-j\sigma}| \bar{A}_j / (j^2 |\sigma|)$  and  $R_1 \leq B_N(1 + \log(\pi n/y)) + O(1)$ , with  $B_N = 2 + 1.6c_1$  for  $h \neq 0$  and  $B_N = 2 + 4c_1$  for  $h = 0$ .
- *Part II*,  $j > \pi n/y$ . Split the  $j$  into periods  $k \geq 1$ , with  $jy/n \in ((2k-1)\pi, (2k+1)\pi]$ . Write  $a = jL/n$  and  $b = jy/n$ , so that  $j\varsigma = a - ib$ . On period  $k$ :  $1/j \leq y/((2k-1)\pi n)$ ;  $|1 - e^{-j\varsigma}| \geq \max(1 - e^{-a}, 2e^{-a/2} |\sin(b/2)|)$ ;  $|\sin(b/2)| \geq |b - 2\pi k|/\pi$  on the period. The two  $j$  nearest the pole  $b = 2\pi k$  are bounded by  $G_k = 1/(1 - e^{-a_k^-})$ , where  $a_k^-$  is the smallest  $a$  on the period. The other  $j$  on the period are bounded through  $|\sin(b/2)|$  and summed with  $\sum_{i \leq M'} \min(G, D/i) \leq \min(M'G, D(1 + \log^+(2M'G/D)))$ . Consequently: summed over  $k$ , the pole terms total at most  $\beta u^2 n/L$ , with  $u = y/n$  and  $\beta = 1 + e^{-\pi L/u}$ ; the remaining terms of the periods total  $O(\log^2 n)$ ; the periods with  $a_k^- > 2$  give a constant tail, bounded with  $\int_2^\infty \phi(a)/a da$  in closed form. Part II is non-increasing in  $n$  at fixed  $(L, y)$ :  $M'/n$ ,  $D/n$  and  $\log^+(2M'G/D)$  do not increase, and  $\beta_k$  decreases. (N) stops at  $u = 0.25$  because the pole term  $\beta u^2/L$  must stay below  $\Lambda(L) - 0.1$  at  $L = 0.1$ . The triangle-bound majorant blows up at low-denominator rationals such as  $\theta = \pi$  and  $2\pi/3$ , and that is where (F) takes over.

The certificate has two parts: **(N-a)**  $y \in [Y_E, 200]$  on  $(L, y)$ -boxes, at  $N_0$ , with the exact  $M_N(\sigma)$  (400 terms plus a tail) and slope check  $g \geq (B_N + 1.5)/N_0$  (monotonicity in  $n$  follows from  $R_1 = A + B_N \log n$ , Part II being non-increasing, and the slope condition); **(N-b)**  $y \in [200, 0.25n]$  on  $(L, u)$ -boxes, in the form  $-nG + a + b \log n + c \log^2 n$  with  $G \geq (b + 2c \log N_0)/N_0$ . The worst values are  $-147.9/-23.3$  for  $h = 1$  and  $-71.3/-1.8$  for  $h = 0$ . The script runs only  $h \in \{0, 1\}$  with  $y > 0$ . This suffices: the majorant depends on  $h$  only through  $\bar{A}_j$ , which equals 1 if  $5 \nmid j$  and 4 if  $5 \mid j$  for every  $h = 1, \dots, 4$ , and it depends on  $y$  only through  $|1 - e^{-j\sigma}|$ ,  $|\sigma|$  and  $|y|$ , all invariant under  $y \mapsto -y$ .

*Certificate:* `offpeak_tb.py 821 0.25; log third/logs/offpeak_tb_821.log`.

Piece (F):  $\text{dist}(\theta, \{0, 2\pi/5, 4\pi/5\}) \geq 0.0499$ ,  $\theta \leq \pi$ ,  $L \in [0, 2.1]$ . The chunk length is  $T = 40$  on the bands  $0.05 \leq \text{dist} < 0.1$  and  $T = 20$  elsewhere. Put  $M_T = \lfloor n/T \rfloor$ ,  $\lambda_T = TL/n$ ,  $\mu_r = L/(5n)$  and  $\Pi(Z) = \prod_{k' \leq 5T, 5 \nmid k'} (1 - Ze^{ik'\tilde{\theta}})$  with  $\tilde{\theta} = \theta + i\mu_r$ . Here the radial drift  $r^{k'}$  appears as an imaginary shift of  $\theta$ .

**Lemma 4.31** (chunk lemma).

$$\log |P_n| \leq \sum_{m' < M_T} \mathcal{B}(m'\lambda_T) + 4(T-1) \log(1 + e^{-(L-\lambda_T)}), \quad \mathcal{B}(s) = \max_{|Z|=e^{-s}} \log |\Pi(Z)|,$$

$$n\Lambda(L) \geq \sum_{m' < M_T} TQ(m'\lambda_T) - T[\lambda_T + (\log 5 - Q(L))/2].$$

*Proof.* Write  $n = M_T T + r_T$  with  $0 \leq r_T < T$ . For  $m' < M_T$ , the factors with  $5Tm' < k \leq 5T(m'+1)$  form one chunk: with  $k = 5Tm' + k'$  and  $Z_{m'} = q^{5Tm'}$ , the chunk equals  $\Pi(Z_{m'})$  with  $|Z_{m'}| = e^{-m'\lambda_T}$ , so its log-modulus is at most  $\mathcal{B}(m'\lambda_T)$ . This holds for every  $n$ , whether or not  $T \mid n$ . The factors with  $5TM_T < k \leq 5n$  and  $5 \nmid k$  are left over; there are  $4r_T \leq 4(T-1)$  of them. Each satisfies  $|1 - q^k| \leq 1 + e^{-kL/(5n)}$ , and  $kL/(5n) > TM_T L/n > L - \lambda_T$  because  $TM_T > n - T$ . So the leftover factors cost at most  $4(T-1) \log(1 + e^{-(L-\lambda_T)})$ . The second line uses the tangent line of  $Q$  at each  $m'\lambda_T$ , convexity telescoping  $-Q'(m'\lambda_T)\lambda_T \leq Q((m'-1)\lambda_T) - Q(m'\lambda_T)$ ,  $|Q'| \leq 2$  for the  $m' = 0$  term,  $Q > 0$ , and  $M_T\lambda_T \leq L$ .  $\square$

*Reductions.*  $\mathcal{B}$  is convex and non-increasing in  $s$  (Hadamard three circles). So on an  $s$ -grid of step 0.1,  $\mathcal{B}$  is below its chord and  $TQ$  is above its tangent, which gives cell lower bounds  $d_i$ .  $\sup_Z \log |\Pi|$  is subharmonic in  $\tilde{\theta}$ , since  $\Pi$  is entire in  $\tilde{\theta}$ ; so on each rectangle  $[\theta_a, \theta_b] \times [0, \mu_{\max}]$ , with  $\mu_{\max} = 2.1/(5N_0)$ , the maximum is attained on the boundary, and the drift needs no interior treatment. The assembly uses left Riemann sums of the running minimum of  $d_i$ ; for small  $L$  it uses  $M_T d_{\min}$  instead. It is checked on the  $L$ -intervals  $[i/100, (i+1)/100]$ ,  $i = 0, \dots, 209$ , which cover  $[0, 2.1]$  including  $L = 0$ , at  $N_0$ , with the slope condition, so it holds for all  $n \geq N_0$ .

**Lemma 4.32** (piece (F)). *On 163 rectangles and 22  $s$ -values, in Arb ball arithmetic over dyadic cells, the assembled margin is positive on every rectangle; the minimum is 47.91 (piece 7, rectangle 26, ending at  $\pi^+$ ,  $T = 20$ ). The tightest  $T = 40$  band has margin 51.8.*

Certificate: `offpeak_chunk_arb.py 821 $p` ( $p=0, \dots, 7$ ), then `offpeak_chunk_arb.py summarize; log third/logs/arb_F-summary-full.log` (merging the 13 logs `arb_F*p*.log`).

The same bound was first certified in float64 interval arithmetic with directed-rounding margins (only correctly rounded  $+$ ,  $-$ ,  $\times$ ,  $\div$  and exact `frexpldexp`; transcendental inputs from Arb with explicit margins), with identical minimum margin 47.9 (`offpeak_chunk.py 821 all, log offpeak_chunk.821.log`); the Arb certificate supersedes it and also fixes the endpoint  $\pi$  and an inexact key in the assembly. An early exit in the Arb cell test that could only reject cells, never accept them, stalled the first runs; it was removed, and every rectangle logged before the fix remains valid (an audit re-ran 11 of them, including the worst, with identical margins).

*Coverage of the  $\theta$ -circle.* The pieces meet or overlap at every handover: (A)–(E) abut at  $|y| = 1.1$  (the statement has a strict  $>$ ); (E)–(N) abut at  $|y| = 6$ ; (E0)–(N) abut at  $y = 20$ ; (N-a)–(N-b) meet at  $y = 200$ , which needs  $0.25n \geq 200$ , i.e.  $n \geq 800$ ; (N)–(F): (N) reaches  $|\theta - 2\pi h/5| = 0.25n/(5n) = 0.05$  for every  $n$ , and (F) starts at 0.0499, an overlap of  $10^{-4}$  rad. The root  $6\pi/5$  is 0.63 away from  $\pi$ , so  $[0, \pi]$  needs no exclusion around it. The (F) rectangles tile each piece exactly, and the pieces abut exactly. The float-endpoint slivers are closed by the closure remark. This proves Lemma 4.29.

*Spot validation (not part of the proof).* 312 random points and, in a second audit, 5570 adversarial points (for  $n$  from 821 to  $10^5$ , window edges, box endpoints,  $L$  down to  $10^{-9}$ , located

local maxima) were evaluated with exact Arb full products: 0 violations; the smallest true slack is 43.18, at  $n = 821$ ,  $L = 2.1$ ,  $|y| = 1.1^+$ .

4.6.3. *Combination.* On the off-peak set, Lemma 4.29 gives  $|P_n r^{-m}| \leq e^{n\Lambda(L)+2n\mu L-\mathcal{E}(n)}$ . The set has measure at most  $2\pi$ , so

$$|\text{OFF}| \leq \frac{1}{\mathcal{N}} \cdot \frac{2\pi}{2\pi} e^{n\Lambda+2n\mu L-\mathcal{E}(n)} = 10\pi n \sqrt{\frac{na_L}{2\pi}} e^{-\mathcal{E}(n)} = 10\pi \sqrt{\frac{a_L}{2\pi}} e^{-6.3}.$$

The powers of  $n$  cancel exactly, so this bound is independent of  $n$ . It grows with  $a_L = \Lambda''(L)$ . The largest  $a_L$  occurs at  $L = 0$ , where  $a_L = 2/3$ , while the minimum slack of Lemma 4.28 occurs at  $L \approx 2.1$ ; so the comparison must be made per  $L$ -box.

**Lemma 4.33** (combination). *On every one of the 210  $L$ -boxes and for every class  $s$ , with  $\text{OFF}_{up} = 10\pi \sqrt{a_{up}/2\pi} e^{-6.3}$  computed in Arb with the exact constant `arb('6.3')`, the program asserts  $M_{s,lo} - \text{err}_{s,up} - \text{OFF}_{up} > 0$  in Arb. Over all boxes and classes the minimum of slack minus OFF is 0.010043, on the box  $[2.09, 2.10]$  (slack 0.018703, OFF 0.008660); on the last box actually needed,  $[2.08, 2.09]$ , it is 0.0120 (0.020699 – 0.008696).*

Certificate: `centre3_comb.py 821 210 64 0 105; centre3_comb.py 821 210 64 105 210;`  
`log third/logs/centre3_comb-821.log.`

**Corollary 4.34.** *For  $n \geq 821$ ,  $\mu \in [0.105, \frac{1}{2}]$  and every class,  $c(m)$  has the conjectured sign.*

*Proof.* Lemmas 4.27, 4.28, 4.29 and 4.33. □

*Validation (not part of the proof).* An audit evaluated  $c(m)/\mathcal{N}$  by quadrature on the windows, validated against exact coefficients at  $n = 300$  (35 points, agreement to  $4 \cdot 10^{-11}$ ), and then checked 120 points at  $n = 821$ , 857 and 900, with  $\mu \in \{0.105, 0.15, 0.2, 0.3, 0.4, 0.45, 0.49, 0.5\}$  and all five classes: every sign correct; realised error at most  $1.65 \cdot 10^{-3}$  (at most  $4.7 \cdot 10^{-4}$  in classes 3 and 4), against a certified err of 0.06–0.21.

#### 4.7. Coverage and proof of Theorem 4.1.

**Theorem 4.35** (coverage). *Let  $n \geq 980$  and  $0 \leq m \leq 5n^2$ . Then  $(n, m)$  is covered by at least one of the following: the band certificates of §4.4 (if  $\lambda < 4$ ); the Laplace certificates of §4.5 (if  $\lambda \geq 4$  and  $\mu < 0.105$ ); the centre certificate of §4.6 (if  $\mu \geq 0.105$ ). Each certificate that applies proves the conjectured sign of  $c_n(m)$ .*

*Proof.* The three conditions partition  $\{0 \leq m \leq 5n^2\}$ . ( $\lambda \geq 4$  is automatic when  $\mu \geq 0.105$ , since  $m \geq 1.05n^2 > 4(5n+1)$  for  $n \geq 22$ , but it is not needed.) It remains to check that each certificate's hypotheses hold on its whole region.

- (1) *Band*,  $m < 4N$ . Classes 3, 4: Corollary 4.15 covers  $m < 2N$  for every  $n$ ; for  $2N \leq m < 4N$ , the exact computation covers  $n \leq 2000$  and  $\Psi$  covers  $n > 2000$ . Classes 0, 1, 2: the exact computation covers  $n \leq 2000$ , and Lemma 4.17 covers  $n > 2000$ . The only threshold is  $n \geq 291$  for the exact band run, which holds.
- (2) *Laplace*,  $\lambda \geq 4$ ,  $\mu < 0.105$ . By §4.5.5 the saddle equation, for either class family, has a root, and every root satisfies  $V > 2$ . Fix a root. Then  $W = V/(5n)$ , the representation of §4.5.1 is exact, and  $X = 5n/V$ . If  $V \in [2, 12]$ ,  $V$  lies in one of the 500 tiles  $[V_{lo}, V_{hi}]$ ; for both class families the tiles are contiguous and the last tile is snapped to end exactly at 12. Classes 0–2:  $X \geq 1455/V_{hi}$ , since  $n \geq 291$ , and  $X \geq 4V_{lo}/a_5$ , since  $\lambda \geq 4$ ; so  $X \geq X_{\min}$  of the tile. Classes 3–4:  $X \geq 4105/V_{hi}$ , since  $n \geq 821$ , and  $X > 3V_{lo}/a_5$ , since  $\lambda \geq 4$ . If  $V \geq 11.99$ , the tail cell applies; for both class families it is run from  $V_T = 11.99$ , so it overlaps the last tile. It needs  $X \geq 4V/a_5$  (classes 0–2) or  $X \geq X^* = 216.3$  together with  $V \leq a_5 X/3$  (classes 3–4), and these follow from the same two inequalities. Every certificate term is non-increasing in

$X$ , so the tile's certified inequality holds at this  $X$ . The thresholds are  $n \geq 291$  (classes 0–2) and  $n \geq 821$  (classes 3–4), both below 980.

- (3) *Centre*,  $\mu \geq 0.105$ . Since  $m \leq 5n^2$ ,  $\mu \leq \frac{1}{2}$ . By §4.6.1,  $\mu \mapsto L$  is a bijection from  $[0.105, \frac{1}{2}]$  onto  $[0, L(0.105)] \subset [0, 2.09]$  (since  $L(0.105) = 2.0814$ ), and every  $L$ -box in  $[0, 2.1]$  is certified at  $N_0 = 821$  for all  $n \geq 821$ . The integration circle is split into the four near-peak windows  $|y_h| \leq 1.1$  and their complement. The complement is covered by (E), (E0), (N) and (F), with the handovers listed in §4.6.2 and the closure remark for the float endpoints, and Lemma 4.29 holds there. The per-box combination of Lemma 4.33 then gives the sign. The threshold is  $n \geq 821$ .

So all thresholds (291, 821, and 2000 for the switch between exact and analytic in the band) are satisfied or bridged for  $n \geq 980$ . The boundaries  $\lambda = 4$  and  $\mu = 0.105$  are assigned to the Laplace region and the centre respectively. Each certificate includes its closed boundary: the Laplace certificates use  $\lambda \geq 4$ , and the centre uses  $\mu \geq 0.105$  through  $\mu(2.1) < 0.105$ . So nothing is left out.  $\square$

*Proof of Theorem 4.1.* By Lemma 2.2 it suffices to take  $0 \leq m \leq 5n^2$ . For  $n \leq 979$  the result is Theorem 4.11. For  $n \geq 980$  it follows from Theorem 4.35, together with Corollaries 4.16 and 4.18, Lemmas 4.20, 4.21 and 4.23, and Corollary 4.34. The analytic inputs are Theorem 4.3, the elementary lemmas proved above, and the standard facts of Appendix B.  $\square$

## 5. COMPUTATIONAL VERIFICATION

All certificates are Python programs using FLINT [12] and Arb [7] through `python-flint`. They are published with the paper [8] in two directories, `cubic/` (Theorem A) and `third/` (Theorem B), each with its programs in `certificates/`, its logs in `logs/`, and a script `run_all.sh` that regenerates every log. A certificate consists of a program, the log it writes, and a line that the log must contain; every program aborts on a failed assertion. A printed bound is an upper (or lower) end of a certified enclosure.

**5.1. Certificates for Theorem A.** Programs in `cubic/certificates/`, logs in `cubic/logs/`. The whole set is regenerated by `cubic/run_all.sh` in about three hours on one core; the uniform Laplace grid parallelises over its 36 chunks (`JOBS=16 ./run_all.sh`).

| certifies                                                        | program(s)                                               | log: required line                                                                            |
|------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| finite range $n \leq 243$<br>(Cert. 3.9)                         | <code>exact_small_n.py</code> 1<br>200; 201 225; 226 243 | <code>01_exact_small_n_{a,b,c}.log</code> : EXACT $n$ in<br>[.]: violations 0 in all three    |
| band identity and<br>$g(i) < 0$ , $i \leq 36000$<br>(Cert. 3.11) | <code>band_lemma.py</code> 36000                         | <code>02_band_lemma.log</code> : (a) ... True, (b) ...<br>True                                |
| band tail $i \geq 36000$<br>(Prop. 3.10)                         | <code>band_tail_rigorous.py</code>                       | <code>02b_band_tail.log</code> : BAND TAIL: $g(i) < 0$<br>for all $i \geq 36000$              |
| constants of Thm 3.4                                             | <code>sz_expansion_audit.py</code>                       | <code>03_sz_audit.log</code> : $E_{\text{con}}=50.150108$                                     |
| Poisson bound (Cert. 3.6)                                        | <code>poisson_check.py</code>                            | <code>04a_poisson.log</code> : $\max  \text{eps}  \dots \leq$<br>1.420                        |
| uniform Laplace grid<br>(Cert. 3.14)                             | <code>run_uniform.py</code> (36<br>chunks)               | <code>uni_*.log</code> : every line UNRESOLVED 0; worst<br>$B_{\text{tot}} = 0.1169$          |
| closing lemma<br>(Prop. 3.15)                                    | <code>ray_asymptotic2.py</code>                          | <code>07_closing_lemma.log</code> : CLOSING LEMMA ( $v$<br>$\geq 3000$ , all $N$ ): CERTIFIED |

| certifies                                        | program(s)                                                                            | log: required line                                    |
|--------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------|
| centre tables (Cert. 3.17)                       | centre_mid_uniform{,2,3}.py,<br>centre_g3_uniform{,2,3}.py,<br>centre_clow_uniform.py | *_table.txt (115, 115, 74 lines): no assertion errors |
| coverage and centre combination (Prop. 3.18)     | coverage.py                                                                           | 08_coverage.log: PASS A-PASS G, COVERAGE COMPLETE     |
| <i>Consistency checks, not part of the proof</i> |                                                                                       |                                                       |
| expansion bound, $i \leq 108000$                 | expansion_check.py<br>36000                                                           | 03b_expansion_check.log: max  rest /bound = 0.8660    |
| Appendix B.2 checks                              | hand_checks.py,<br>koksma_cauchy_check.py                                             | 04b, 04c: ratios $\leq 1$                             |
| Prop. 3.7 end to end                             | identity_check.py                                                                     | 04d_identity.log: rel. err. $< 10^{-8}$ on every line |

expansion\_check.py uses mpmath floating point with adaptive precision; it corroborates §3.6 and is not part of the proof.

**5.2. Certificates for Theorem B.** Programs in third/certificates/ (together with the shared modules offpeak\_common.py, cls012\_lib.py, lap34\_lib.py and the exact coefficient table g5\_coeffs.pkl), logs in third/logs/. Runtimes are those printed in the logs or recorded by the audits, on a 2-core machine.

| certifies                                            | command                      | log: required line                                                                                       | time       |
|------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------|------------|
| exact, $1 \leq n \leq 150$<br>(Thm 4.11)             | centre3_exact.py 1 150       | exact_1_150.log: DONE n=1..150<br>total violations 0                                                     | 4s         |
| exact, $150 \leq n \leq 291$                         | centre3_exact.py 150 291     | exact_150_291.log: DONE n=150..291<br>total violations 0                                                 | 34s        |
| exact, $291 \leq n \leq 410$                         | centre3_exact.py 291 410     | centre3_exact_291_410.log: DONE ...<br>violations 0                                                      | 114s       |
| exact, $410 \leq n \leq 580$                         | centre3_exact.py 410 580     | centre3_exact_410_580.log: DONE ...<br>violations 0                                                      | 419s       |
| exact, $580 \leq n \leq 820$                         | centre3_exact.py 580 820     | centre3_exact_580_820.log: DONE ...<br>violations 0                                                      | 1357s      |
| exact, $820 \leq n \leq 920$                         | centre3_exact.py 820 920     | centre3_exact_820_920.log: DONE ...<br>violations 0                                                      | 2825s      |
| exact, $920 \leq n \leq 979$                         | centre3_exact.py 920<br>1000 | centre3_exact_920_1000.log: 60 lines<br>n=920 ... n=979 wrong_sign=0, then<br>Killed (about 6 GB needed) | 3786s      |
| band, exact,<br>$291 \leq n \leq 2000$ ,<br>$m < 4N$ | bandfix_exact.py 291<br>2000 | bandfix_exact.log: sign failures<br>(any class, $m < 4N$ ): 0                                            | 192s       |
| coefficient table;<br>Lemma 4.14                     | bandfix_coeffs.py            | bandfix_coeffs.log: g5_coeffs.pkl<br>agrees ... and ONE-PART BLOCK<br>(exact): ...                       | <<br>1 min |

| certifies                                           | command                                                                      | log: required line                                                                                                           | time             |
|-----------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------|
| Lemma 4.13, $\Psi$                                  | bandfix_analytic.py<br>10001                                                 | bandfix_analytic.log: L1: ... <= <<br>[0.017972 +/- 1.12e-7]; BAND<br>(analytic): ... < 0.1977 for<br>all N >= 10001         | 1 min            |
| band classes 0–2<br>(Lemma 4.17)                    | cls012_band.py                                                               | cls012_band.log: PART A: ... sign<br>failures 0; PART B: ... Q <=<br>[5.1390e-20 ...]                                        | seconds          |
| Laplace 0–2, tiles<br>(Lemma 4.20)                  | cls012_lap.py 2.0 12.0<br>0.02                                               | cls012_lap_rerun.log l. 1: 500<br>tiles, 0 failures; ... 0:<br>3.0700, 1: 1.6879, 2: 0.8339                                  | 8 s              |
| Laplace 0–2, tail<br>(Lemma 4.21)                   | cls012_lap.py tail 11.99                                                     | cls012_lap_rerun.log l. 2: tail cell<br>V >= 11.99 : ({0: (3.0950...                                                         | seconds          |
| saddle coverage<br>(Lemma 4.24)                     | cls012_cov.py                                                                | cls012_cov.log: 5858 tiles, min A<br>= [0.04424...]                                                                          | seconds          |
| Laplace 3–4<br>(Lemma 4.23)                         | lap34_cert.py 2.0 12.0<br>0.02; lap34_cert.py tail<br>11.99                  | lap34_cert.log: 500 tiles, 0<br>failures; margins 0.8363, 0.8361;<br>tail cell V >= 11.99 : ({3:<br>(0.8145634857582285, ... | 76 s             |
| saddle endpoint<br>(Lemma 4.27)                     | centre3_saddle_end.py                                                        | centre3_saddle_end.log: ... <<br>0.105: True                                                                                 | seconds          |
| near peak and<br>combination<br>(Lemmas 4.28, 4.33) | centre3_comb.py 821 210<br>64 0 105 and ... 105<br>210                       | centre3_comb.821.log: ALL_OK on<br>every box; twice COMBINATION<br>ASSERTED (slack > OFF, arb) on<br>every box; ... True     | $\approx 11$ min |
| off-peak (E)                                        | offpeak_em.py 821 6.0 32                                                     | offpeak_em.821.log: piece (E)<br>CERTIFIED for all n >= N0                                                                   | 57 s             |
| off-peak (E0)                                       | offpeak_em0.py 821 20 32                                                     | offpeak_em0.821.log: piece (E0)<br>CERTIFIED for all n >= N0                                                                 | 6 s              |
| off-peak (N)                                        | offpeak_tb.py 821 0.25                                                       | offpeak_tb.821.log: piece (N)<br>CERTIFIED for all n >= N0                                                                   | 2 s              |
| off-peak (F), Arb<br>(Lemma 4.32)                   | offpeak_chunk_arb.py 821<br>\$p -j 2, p = 0, \dots, 7; then<br>... summarize | arb_F_summary_full.log: (F) arb:<br>CERTIFIED for all n >= N0 (all<br>rects OK)                                              | $\approx 1.5$ h  |
| off-peak (F), float<br>(superseded)                 | offpeak_chunk.py 821 all                                                     | offpeak_chunk.821.log: (F) all:<br>CERTIFIED                                                                                 | 470 s            |

centre3\_peak.821.log (centre3\_peak.py 821 210 64) contains the near-peak part alone and is reproduced inside centre3\_comb.821.log. The coefficient table g5\_coefs.pkl is produced by g5\_coef.py. The file arb\_F\_summary.log of the working directory, an incomplete earlier summary, is not a certificate and is not shipped.

### 5.3. Exact ranges.

- Theorem A:  $P_{3,n}^3$  exactly for  $n \leq 243$ ; exact  $s(i)$  and  $g(i) < 0$  for  $i \leq 36000$ ; the band identity for  $n \leq 60$ .
- Theorem B:  $P_{5,n}$  exactly for  $n \leq 979$  (all classes, all  $m$ ); the band  $m < 4N$  exactly for  $291 \leq n \leq 2000$ ; exact  $s(i)$  for  $i \leq 200000$ ; the one-part block for  $t \leq 39999$ ; Lemma 4.17(A) for  $m \leq 200000$ .

**5.4. Reproducibility.** The software versions are Python 3.12, `python-flint` 0.9.0, `mpmath` 1.3.0, `numpy` 2.4.4, `scipy` 1.17.1 and `sympy` 1.14.0. The repository’s top-level `Makefile` has targets `verify-cubic` and `verify-third`, which call the two `run_all.sh` scripts. The expensive steps are the cubic uniform Laplace grid and exact range (about three hours on one core in total), and, for Theorem B, the exact range  $n \leq 979$  (about 2.4 hours of CPU time, and about 6 GB of memory for the last segment), the near-peak and combination grid (about 11 minutes per half), and the Arb run of piece (F) (about 1.5 hours on two cores). All other certificates run in seconds to a few minutes.

**5.5. Trust base.** The proofs rely on:

- (1) the written arguments of Sections 2–4;
- (2) the standard facts listed in Appendix B;
- (3) the correctness of Arb (every ball returned contains the true value) and of FLINT’s exact integer polynomial arithmetic, as accessed through `python-flint`;
- (4) the correctness of the Python interpreter executing the certificate programs as written.

IEEE-754 floating point enters only in the superseded float-interval version of piece (F), and `mpmath` enters only in consistency checks and in the independent recomputation of the constants of Theorem 4.3 (which were computed at 60 digits and independently at 50 digits).

## APPENDIX A. VERIFICATION AND AUDIT RECORD

This appendix records, in concise form, how the proof of Theorem B was checked independently, which defects were found, and which earlier components were withdrawn. The full reports are in `third/verification/` of the repository (`REFeree_SEC3.md`, `REFeree_FULL.md`, `AUDIT*.md`, `AUDIT2_*.md`, `GAP_*.md`, `ARB_F_RUN.md`). The proof of Theorem A is documented by its certificates and consistency checks (§5).

### A.1. Independent checks. *First round.*

- §4.2 (`REFeree_SEC3.md`): independent re-derivation; own `mpmath` code at 50 digits; 10 cusps  $\times$  2 points for (6)/(7); quadrature for Lemma 4.10; exhaustive test  $\nu \leq 5000$  plus 1509 values up to 200000. Verdict: correct; three presentational gaps (the passage from the Hankel loop to the vertical line, the arc of 0/1, the regions of deformation), all filled in §4.2.
- *Band and Laplace, classes 0–2, and coverage* (`GAP_AUDIT012.md`): re-derivation, NaN-guarded rerun of all tiles, 45 independent exact points. Sound; two latent flaws fixed.
- *Band, classes 3–4*: an earlier audit’s findings are repaired in `GAP_BAND.md`.
- *Laplace, classes 3–4* (`AUDIT_LAP34.md`): re-derivation, 6000-point test of  $\varepsilon_g$ , NaN-guarded rerun, 36 exact points with independent code. Sound after two fixes.
- *Near peak and combination* (`AUDIT_CENTRE.md`): re-derivation of the representation, Euler–Maclaurin constants, uniformity in  $n$ , saddle and exact-run logs; exact  $P_{5,300}$  comparison. Sound after replacing the combination step.
- *Off-peak lemma* (`AUDIT_OFFPEAK.md`): coverage in  $\theta$ , uniformity in  $n$ , Lemma 4.30, chunk lemma and maximum principle, 312 exact spot points; Arb version of (F) on 7 rectangles. Sound after cosmetic fixes.

*Second round* (each auditor wrote their findings before reading the first-round report).

- *Band, classes 3–4* (`AUDIT_BAND.md`): exact run re-run; 16 values of  $n$  recomputed by a direct product avoiding  $E_n$  and the coefficient table; pair and triple counts brute-forced; L1 and  $\Psi$  recomputed in `mpmath` ( $\Psi(10001) = 0.19714$ ). Two cosmetic misstatements. Sound.
- *Laplace, classes 3–4* (`AUDIT2_LAP34.md`): re-derivation from scratch; complete term inventory including the constants of  $T_\rho$  against Theorem 4.3; monotonicity in  $X$  at 6 levels; a third

exact method (Euler identities modulo primes, CRT), 31 points up to  $n = 5000$ . No hole. Sound.

- *Centre* (AUDIT2\_CENTRE.md): Cauchy decomposition and  $\mathcal{N}$  re-derived;  $\mu(2.1) = 0.103772$ ; the thin box recomputed in independent Arb code; monotonicity in  $n$  up to  $10^5$ ; quadrature validated to  $4 \cdot 10^{-11}$ , then 120 points. Findings: the comparison slack  $> \text{OFF}$  was printed but not asserted (now asserted, Lemma 4.33); the binding box  $[2.09, 2.10]$  is not needed. Sound.
- *Off-peak* (AUDIT2\_OFFPEAK.md): all ingredients re-derived; (N) compared with the exact  $j$ -series at 70 points; certificates re-run at  $N_0 = 2000, 10^4, 10^5$ ; every (F) rectangle matched; 11 pre-fix (F) rectangles re-run; 5570 adversarial Arb points. Findings: float slivers at  $|y| = 1.1$  and  $L = 0.1$ ; the constant 6.3 used as a float,  $1.8 \cdot 10^{-16}$  low. All fixed (§4.6.2). Sound.
- *Classes 0–2* (AUDIT2\_CLASS012.md): representation re-derived, every term inventoried, monotonicity in  $X$  tested up to  $1000X_{\min}$ , 76 hard points validated by an exact CRT method over 82 primes: all signs correct, true error 15–260 $\times$  below the certified bound. No mathematical error.

*Outside referee* (REFEREE\_FULL.md). The whole proof was read end to end, with the certificates treated as black boxes, 13 logs and two scripts spot-checked. Verdict: correct with gaps to fill. One major point: for Laplace classes 0–2 the float-stepped tiling ended at 11.999999999999831 while the tail cell started at 12. This is now closed twice (last tile snapped to 12, tail cell run from 11.99; Lemmas 4.20 and 4.21). Nine minor points were presentational and are addressed.

## A.2. Defects found and fixed in the components used.

- *Expansion*. The Farey order must be  $N = K(\nu)$ , not  $K(\nu) - 1$ ;  $E_{\text{con}}$  is rounded up to 27.4636 (it had been rounded down);  $K_5$  is rounded up to 2.8549955.
- *Band*. An earlier majorant used a Bessel envelope in the wrong direction; an earlier extension certificate undercounted pairs by  $4\times$  and triples by  $10\times$ ; an earlier monotonicity script dropped NaN results silently. All are superseded by the exact run to  $n = 2000$  and by  $\Psi$ ; the threshold for “ $\Psi$  decreasing” is  $N \geq 257$ .
- *Laplace, classes 0–2*. The pass test now rejects NaN (**not** `all(x > 0)`); the decay assertion uses  $\kappa - a_5/4$  (the  $k = 10$  Rankin term) instead of  $\kappa - a_5/5$ ; the  $V = 12$  sliver is closed.
- *Laplace, classes 3–4*. The  $V = 12$  sliver is closed in the same way; NaN pass-through in suprema and in the pass test is guarded; the log is regenerated with the fixed script.
- *Centre*. The scalar comparison  $|\text{OFF}| \leq 0.0187 \leq \text{slack}$  of an earlier version is false (0.018791 against 0.018703); it is replaced by the per-box comparison, now asserted in Arb.
- *Off-peak lemma*. The sliver  $(\text{fl}(\pi), \pi]$ , the  $h/\text{sign}$  reduction for (N), an inexact key in the assembly and the validation of the float checker are fixed or superseded by the Arb run of (F); an early exit in the Arb cell test (which could only reject) was removed; float slivers at  $|y| = 1.1$  and  $L = 0.1$  are closed; 6.3 is an exact ball everywhere.
- *Earlier centre certificate*. The strip next to each peak was bounded by nothing; the tail was normalised with a lower curvature,  $7.4\times$  too small; the tail lemma was unproved (now Lemma 4.29); an Euler–Maclaurin constant was invalid; a Taylor constant was evaluated at the centre instead of over the window; an arctan law was used without its error;  $L < 0.05$  was not covered. All are fixed in the present near-peak analysis.

**A.3. Withdrawn components.** The following earlier components are not used anywhere in this paper.

- `B_uniform5.py`, a 983-tile Laplace certificate for all classes: NaN cells had been replaced by  $e^{-10^6}$ , silently dropping 7168 cells; with a NaN guard, 63 of 73 tiles fail.
- The old Laplace certificate for classes 3–4 (`mono_cert.py`, `zone2_cert.py`, `tail3_cell.py` and related): it was centred on the kernel-only saddle  $m = a_5/W^2$  and treated the  $O(N)$  thin factor  $e^{-5nu}$  and the twisted-product factor as amplitudes. Against the exact line integral

its main term was off by factors down to  $e^{-13}$ , while it certified  $B \leq 0.06$ , so “ $B < 1$ ” proved nothing. Superseded by the re-centred saddle of §4.5.4.

- A folded identity with  $D_0 = 1.12601$ : correct, but unnecessary, since §4.5.1 is exact without folding.
- The old centre certificate: it had the holes listed above; class 4 fails at the handover once the tail normalisation is corrected. Superseded by §4.6.
- Earlier band extensions and a  $t_{\min}$  table with a wrong normalisation.

## APPENDIX B. STANDARD FACTS AND ELEMENTARY LEMMAS

### B.1. Standard facts.

**Fact B.1** (Dedekind eta / partition-function transformation). *This is Fact 4.8, in Rademacher’s normalisation with  $hH \equiv -1 \pmod{k}$  [2, Thm 5.1]. It was confirmed numerically to  $10^{-49}$  at 10 cusps. The form (1) for  $G_p^\delta$  follows from it [11, 9].*

**Fact B.2** (Farey dissection). *The Farey dissection and the Rademacher path via Ford circles (Lemma 4.4); see [2, Ch. 5] and [10, Ch. 14].*

**Fact B.3** (Farey neighbours). *Consecutive fractions  $h/k, h'/k'$  of order  $N$  satisfy  $k + k' \geq N + 1$  [6, Thm 30].*

**Fact B.4** (Bessel contour integral). *The Hankel-loop representation of  $I_1$  [5, 10.9.19], [15, §6.22]. The passage from the loop to the vertical line is proved in §4.2.*

**Fact B.5** (Bessel facts).  *$I_\nu(y) > 0$  for  $y > 0$ , and  $I_\nu(y)$  is decreasing in  $\nu \geq 0$  for fixed  $y > 0$  [4, 15]; the closed forms of  $I_{1/2}$ ,  $I_{3/2}$  and  $I_{5/2}$ ; and  $\mathcal{P}_p(i) = \sqrt{a_p/x} I_1(2\sqrt{a_p x})$ ,  $x = i - B/2$ , is the inverse Laplace transform of  $e^{a_p/u} - 1$ , which follows from the series of  $I_1$ . These are used in §3.2, §4.4 and §4.5.1.*

**Fact B.6** (identities).  *$\sum_{c=0}^4 \text{Li}_2(\omega_5^c x) = \text{Li}_2(x^5)/5$ , with the principal branch for  $|x| < 1$ ; Euler’s pentagonal theorem; Jacobi’s identity for  $(q; q)_\infty^3$ .*

**Fact B.7** (complex analysis). *Cauchy’s theorem, Fubini/Tonelli, the maximum principle for subharmonic functions, and Hadamard’s three-circles theorem (the last two in piece (F)), Poisson summation (Lemma 3.5).*

**Fact B.8** (arithmetic software). *Arb, through `python-flint`: every ball returned contains the true value [7]. FLINT `fmpz_poly`: exact integer arithmetic [12]. IEEE-754 correct rounding of  $+, -, \times, \div$  is relied on only in the superseded float-interval version of piece (F).*

### B.2. Elementary lemmas.

**Lemma B.9** (Cauchy). *If  $f$  is holomorphic on  $|\zeta - c| \leq \rho$  and  $|f - f(c)| \leq M$  on  $|\zeta - c| = \rho$ , then  $|f^{(k)}(z)| \leq k! M \rho / (\rho - d)^{k+1}$  for  $|z - c| \leq d < \rho$ ,  $k \geq 1$ .*

*Proof.*  $f^{(k)}(z) = \frac{k!}{2\pi i} \oint (f(\zeta) - f(c))(\zeta - z)^{-k-1} d\zeta$  and  $|\zeta - z| \geq \rho - d$ . □

**Lemma B.10** (midpoint rule). *For  $g \in C^2[-\frac{1}{2}, \frac{1}{2}]$ ,  $|g(0) - \int g| \leq \frac{1}{8} \int |g''|$ .*

*Proof.*  $g(0) - \int g = -\int K g''$  with  $K(t) = \frac{1}{2}(\frac{1}{2} - |t|)^2 \leq \frac{1}{8}$ . □

**Lemma B.11** (Koksma, one node per cell). *If  $x_i \in [(i-1)h, ih]$  for  $i = 1, \dots, M$ , then  $|h \sum f(x_i) - \int_0^1 f| \leq h \text{TV}(f) + (1 - Mh) \sup |f|$ .*

*Proof.*  $|h f(x_i) - \int_{C_i} f| \leq h \text{TV}_{C_i}(f)$ ; sum, and bound the uncovered part. □

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DEPARTMENT OF COMPUTER SCIENCE, PURDUE UNIVERSITY, WEST LAFAYETTE, IN 47907, USA  
 Email address: [jpadhi@purdue.edu](mailto:jpadhi@purdue.edu)